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CHAPTER

ATA-36

PNEUMATICS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Pneumatics, General

Introduction

Bleed air for the pneumatic systems can be supplied by the engines or by an optional Auxiliary Power Unit (APU).

The APU supplies the system with bleed air prior to engine start. The engines supply bleed air for pneumatics after engine start.

The following systems use pneumatics for operation:

- Air conditioning and pressurization
- Airframe deicing
- Airstair door seal pressurization
- Engine oil cooler ejectors.

APU bleed air, is used for air conditioning on the ground only.

General Description

[Refer Figure 36-1](#)

Bleed air operation from the left and right engines are similar. Compressed Bleed air is ducted from the Low Pressure (LP) compressor port or the High Pressure (HP) compressor port of each engine.

Power lever position and the Electronic Control Unit (ECU) determine whether the HP port or the LP port will be used, when bleeds are selected ON.

The ECU has left and right digital channels and redundant left and right analog channels. The digital channels control the high and low pressure engine bleed ports, monitor protective sensors for temperature, pressure, differential pressure and duct leaks. The digital channels perform a Built-In-Test. The analog channels provide backup control for the digital channels. The analog channels control only LP engine bleed air. The use of HP bleed air and bleed air flow sharing is not permitted under analog control.

Bleed air for airframe deicing, is supplied through the HP or LP bleed air port and passes through the deice section of the precooler. This protects the components of the deice system from overheating. Each digital and analog channel generates a discrete signal to turn on the #1 or #2 BLEED HOT caution light if an overtemperature, overpressure or duct leak occurs. If the digital and analog channel on one side fails, the related BLEED HOT caution light will not turn on.

Selection of the two Bleed Control switches on the AIR CONDITIONING control panel to BLEED 1 and 2, turns on the related engine bleed air system. Two PACKS switches, labelled OFF/MAN/AUTO, and a single rotary BLEED control selector are also on the AIR CONDITIONING control panel. Together they regulate the quantity of air flowing into the system. Selection of the bleed air (BL AIR) switchlight on the APU CONTROL panel turns on the APU bleed air.

If an overtemperature, overpressure or duct leak occurs, the ECU automatically shuts down the bleed air system. The bleed air from each engine flows through a precooler to lower the air temperature before it passes through the wing fuel zones for distribution. The venturis and differential pressure sensors are installed in each wing and APU duct to measure bleed air flow. This information is used by the ECU to control the bleed air flow from each engine or APU.

The ECU uses the NSOV in each nacelle to balance the bleed air output from each engine.

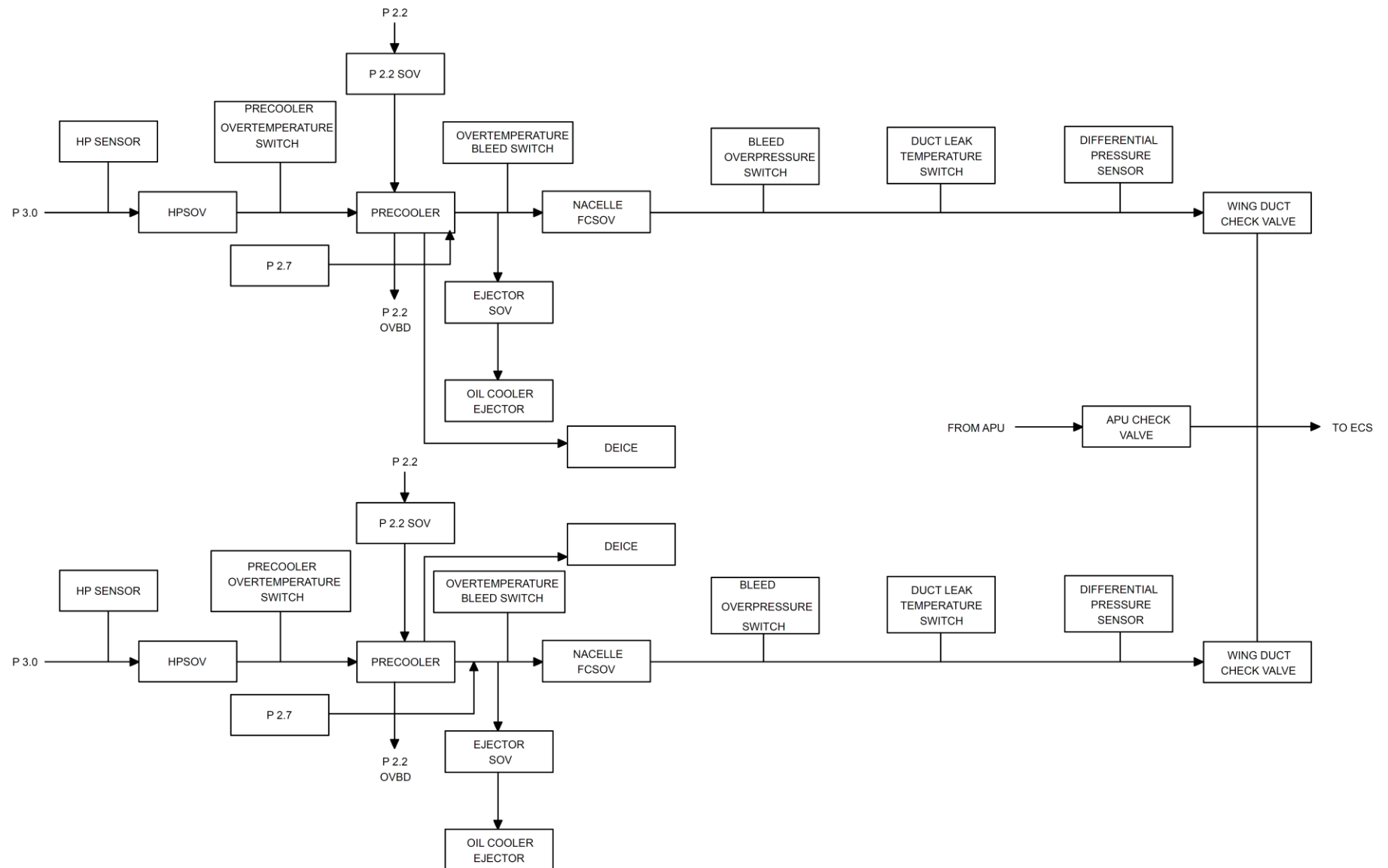


Figure 36-1 Bleed Air System Synoptic Diagram

Bleed Air Switches on Air Conditioning Panel

BLEED 1 AND 2 SWITCHES (two position) Starts bleed airflow from the No. 1 and/or No. 2 engine to the air conditioning packs by enabling operation of the following:

- High pressure shutoff valve
- Nacelle shutoff valve
- P2.2 shutoff valve.

OFF – stops bleed airflow from the selected engine by closing the above mentioned valves.

BLEED CONTROL SELECTOR (three position, rotary action)

MIN – allows Environmental Control System (ECS) controller to modulate the amount of bleed air from both engines at a minimum level. With the BLEED switches on, and NTOP or MTOP set, indicates BLEED (white) on the Engine Display (ED).

NORM, MAX – allows ECS controller to modulate the amount of bleed air from both engines at increased levels. With the BLEED switches on, and NTOP set, indicates BLEED (amber) on the ED. With the BLEED switches on, and MTOP set, rating display changes to, and indicates MCP. BLEED is not displayed.

PACKS CONTROL SWITCHES (three position)

- OFF – shuts related Air Cycle Machine (ACM) off
- MAN – selects analog channel of ECU to supply temperature control based on duct supply temperature
- AUTO – ECU digital channel in control – opens the pack Flow Control and Shut Off Valve – opens turbine Shut Off Valves – ACMs start operating and supply conditioned air to the cabin and flight compartment based on actual cabin temperature.



Bleed Air System

Introduction

The bleed air system supplies compressed air to satisfy the requirements for:

- Air conditioning
- Airframe de-icing
- Door seal pressurization
- Engine oil cooler ejectors
- Cabin and Flight compartment pressurization.

General Description

The Bleed Air System (BAS) is part of the Environmental Control System (ECS) which supplies conditioned air to the flight and cabin compartments of the aircraft.

The BAS has the components that follow:

- Bleed Stage Pressure Switch
- High Pressure Shutoff Valve
- Precooler Temperature Switch
- Precooler
- Overtemperature switch/sensor
- Nacelle Shutoff Valve
- Nacelle Duct Leak Overtemperature Switch
- P2.2 Shutoff Valve
- Overpressure Switch
- Wing Differential Pressure Sensor
- Wing Duct Check Valve
- APU Differential Pressure Sensor
- Aft Fuselage Duct Leak Overtemperature Switches
- APU Check Valve

The BAS has a series of air ducts which transfer bleed air from the engines or the Auxiliary Power Unit (APU) to the Environmental Control System (ECS).

Detailed Description

[Refer Figure 36-2](#), [Refer Figure 36-3](#) & [Refer Figure 36-4](#)

The Bleed Air System (BAS) starts in the nacelles, where it collects compressed air from the engines. In the nacelles, the BAS has valves, heat exchangers, sensors and switches to select, monitor and condition the bleed air drawn from the left and right engines. As commanded by the Electronic Control Unit (ECU), the valves and heat exchangers condition the bleed air for correct temperature, pressure and flow rates. The bleed air is supplied to the ECS, the deice system (including the door seal pressurization system) and the oil cooler ejectors.

Bleed air flows from the two engines through the ducting on each wing forward spar to the wing center section. The wing bleed air ducts, one from each nacelle, join at the wing center section to combine the two engine bleed air flows into one air flow. Each wing bleed air duct has a pressure switch, a differential pressure sensor and a check valve to monitor and control the flow of engine bleed air. The ducts have shrouds and temperature sensors to contain and detect air leaks.

The next section of bleed air ducting transfers the combined engine bleed air flow from the wing center section through the dorsal fin of the fuselage to the tail. This section of air ducts also has a shroud to contain air leaks.

The last section of the BAS is in the aft fuselage. This section transfers the bleed air from the engines or from the Auxiliary Power Unit (APU) to the air conditioning pack. The ducts in this section also have shrouds and temperature switches to contain and detect duct leaks. A differential pressure sensor and a shutoff valve monitor and control bleed air flow from the APU.

The oil cooler ejector shutoff valve (one in each nacelle) controls the flow of engine bleed air for operation of the oil cooler ejector. This valve is controlled by FADEC.

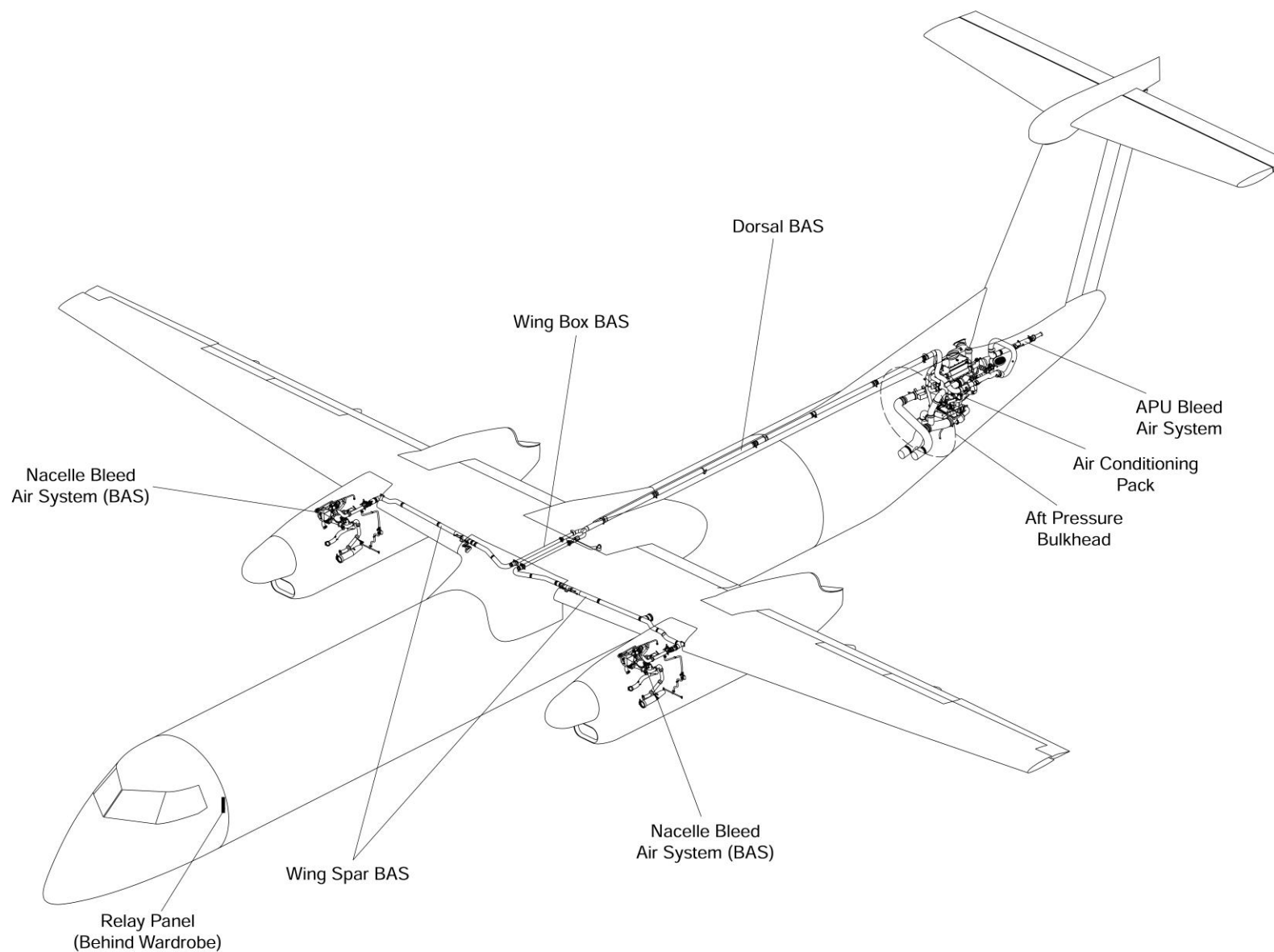


Figure 36-2 Bleed Air System Diagram

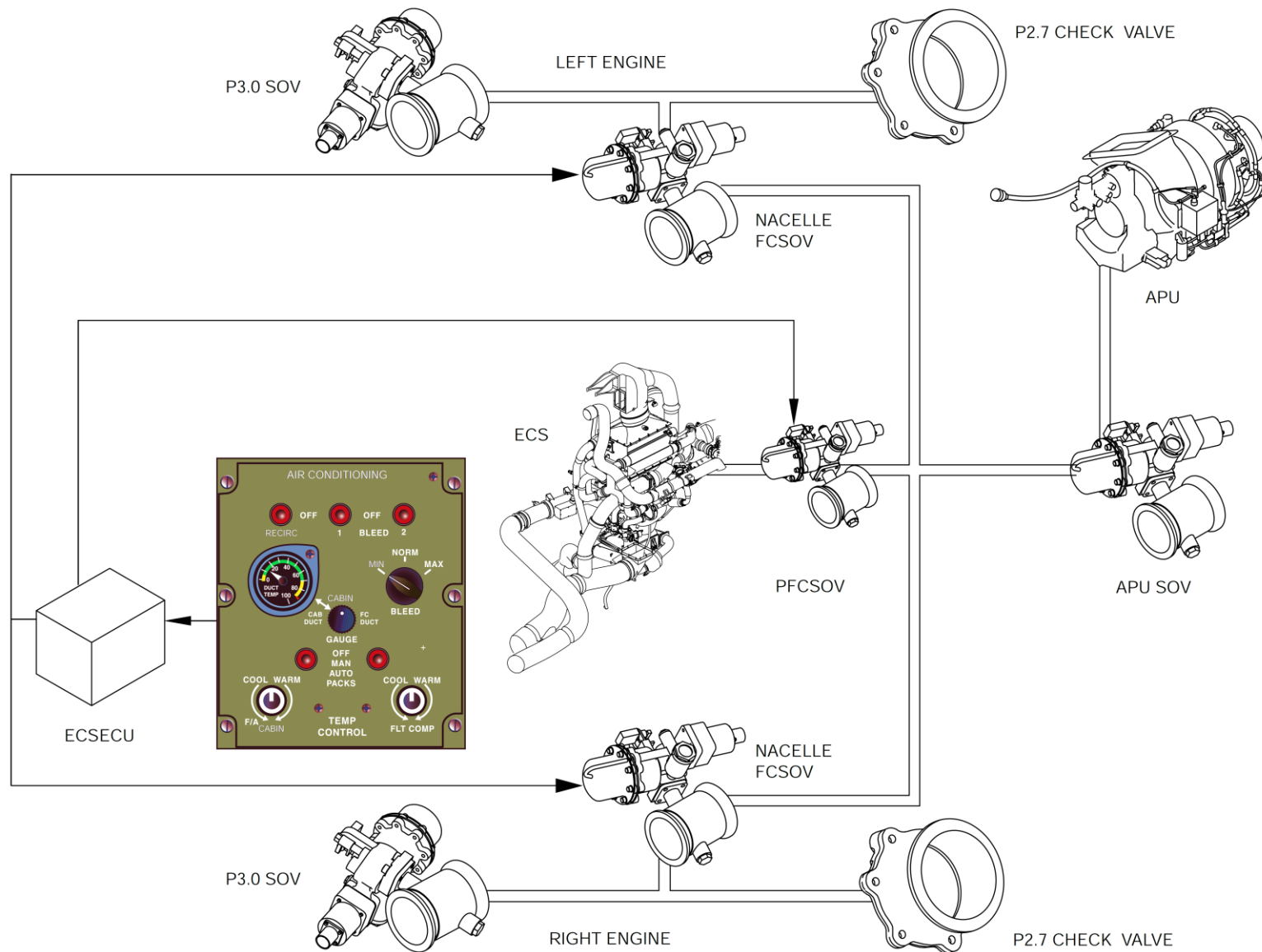


Figure 36-3 Bleed Air System

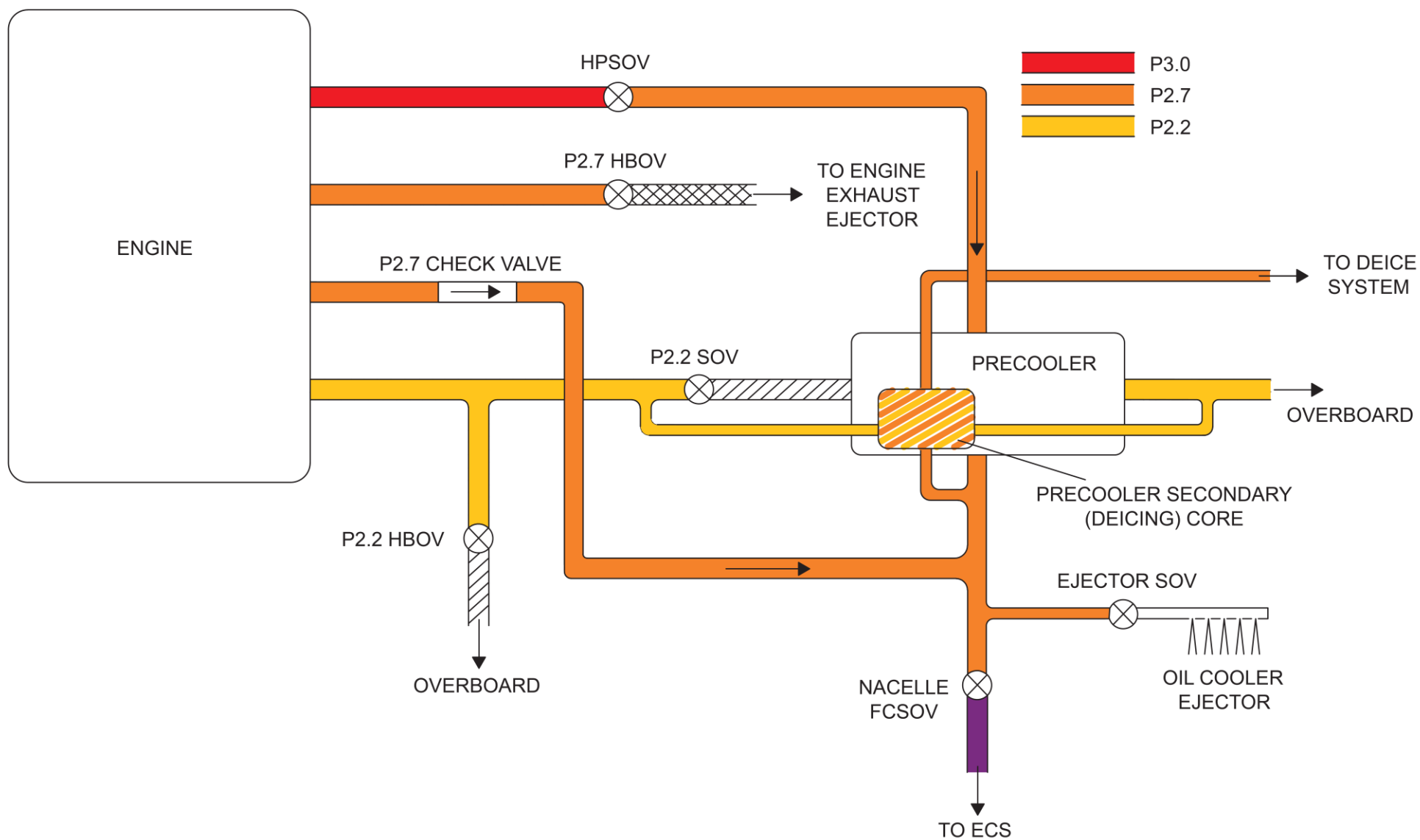


Figure 36-4 Bleed Air System Schematic

The switch settings on the AIR CONDITIONING control panel and the APU CONTROL PANEL determine the selection of bleed air from the engines or the APU.

Refer Figure 36-5

Selection of the BLEED 1 and BLEED 2 switches on the AIR CONDITIONING control panel turns on the left and right engine bleed air systems respectively.

The rotary BLEED MIN/NORM/MAX switch is used to control the amount of bleed air flow. Selection of the BL AIR switch on the APU CONTROL PANEL turns on the APU bleed air system.

Selection of the BLEED 1 and BLEED 2 switches causes the ECU to open both nacelle shutoff valves. This allows bleed air to flow from the left and right engines. Selection of the OFF switch causes the ECU to close the High Pressure Shutoff Valve (HPSOV), P2.2 shutoff valve and the nacelle shutoff valve. This stops the flow of engine bleed air to the ECS.

While bleed air to the ECS is stopped, low pressure (P2.7) bleed air from each engine can flow to the deice system and the oil cooler ejectors. Selection of the APU as a bleed air source by the ECU is done with the APU shutoff valve.

Each engine must be operating to supply bleed air to its oil cooler ejector and to the deice system. Check valves in the wing ducts prevent the use of bleed air from the APU for the deice system and oil cooler ejectors.



Figure 36-5 Air Conditioning and APU Control Panel

System Operation

[Refer Figure 36-6](#) & [Refer Figure 36-7](#)

The switch settings on the AIR CONDITIONING control panel and the APU CONTROL PANEL determine the selection of bleed air from the engines or the APU.

Selection of the BLEED 1 and BLEED 2 switches on the AIR CONDITIONING control panel turns on the left and right engine bleed air systems respectively. The rotary BLEED MIN/NORM/MAX switch is used to control the amount of bleed air flow. Selection of the BL AIR switch on the APU CONTROL PANEL turns on the APU bleed air system.

Selection of the BLEED 1 and BLEED 2 switches causes the ECU to open both nacelle shutoff valves. This allows bleed air to flow from the left and right engines. Selection of the OFF switch causes the ECU to close the High Pressure Shutoff Valve (HPSOV), P2.2 shutoff valve and the nacelle shutoff valve. This stops the flow of engine bleed air to the ECS.

While bleed air to the ECS is stopped, low pressure (P2.7) bleed air from each engine can flow to the deice system and the oil cooler ejectors. Selection of the APU as a bleed air source by the ECU is done with the APU shutoff valve.

Each engine must be operating to supply bleed air to its oil cooler ejector and to the deice system. Check valves in the wing ducts prevent the use of bleed air from the APU for the deice system and oil cooler ejectors.

Bleed air can be supplied from either the high pressure (P3.0) port or the low pressure (P2.7) port on each engine. The ECU uses pressure data from the P3.0 bleed air and environmental data from the Air Data Units (ADUs) to determine which port supplies the bleed air. If the P3.0 bleed air has sufficient pressure, the ECU opens the High Pressure Shutoff Valve (HPSOV).

This allows bleed air from the P3.0 port to supply the Environmental Control System (ECS). A flow-limiting venturi in the high pressure duct limits the

maximum flow of high stage (P3.0) bleed air drawn from each engine. A check valve in each engine's P2.7 bleed duct prevents the back-flow of high pressure P3.0 bleed air into the P2.7 bleed port.

At some engine power settings, the P3.0 bleed air can be too hot (greater than 560 °F (293 °C)) for use by the ECS, deice system and oil cooler ejectors. The BAS uses the high stage precooler (a compact air-to-air heat exchanger in each nacelle) to reduce the P3.0 bleed air temperature. Cooling air from the P2.2 engine bleed port is used in the precooler to lower the temperature of the P3.0 air. The ECU opens the P2.2 shutoff valve to permit the flow of P2.2 bleed air through the high stage precooler.

In the high stage precooler, the P2.2 bleed air absorbs heat from the P3.0 bleed air. The P3.0 bleed air that leaves the precooler is cool enough to flow through the bleed air ducts in the wing fuel zones.

If the initial high stage (P3.0) bleed air temperature is correct, the ECU closes the P2.2 shutoff valve. This disables the high stage precooler operation. This allows P3.0 bleed air to flow directly to the ECS, deice system and oil cooler ejectors with no temperature reduction. The high stage precoolers operate independently on each side of the aircraft.

At higher engine power settings, the ECU closes the HPSOV to select the low stage (P2.7) bleed air. With the HPSOV closed, the P2.7 bleed air pressure opens the P2.7 check valve. This allows P2.7 bleed air to flow directly through to the ECS.

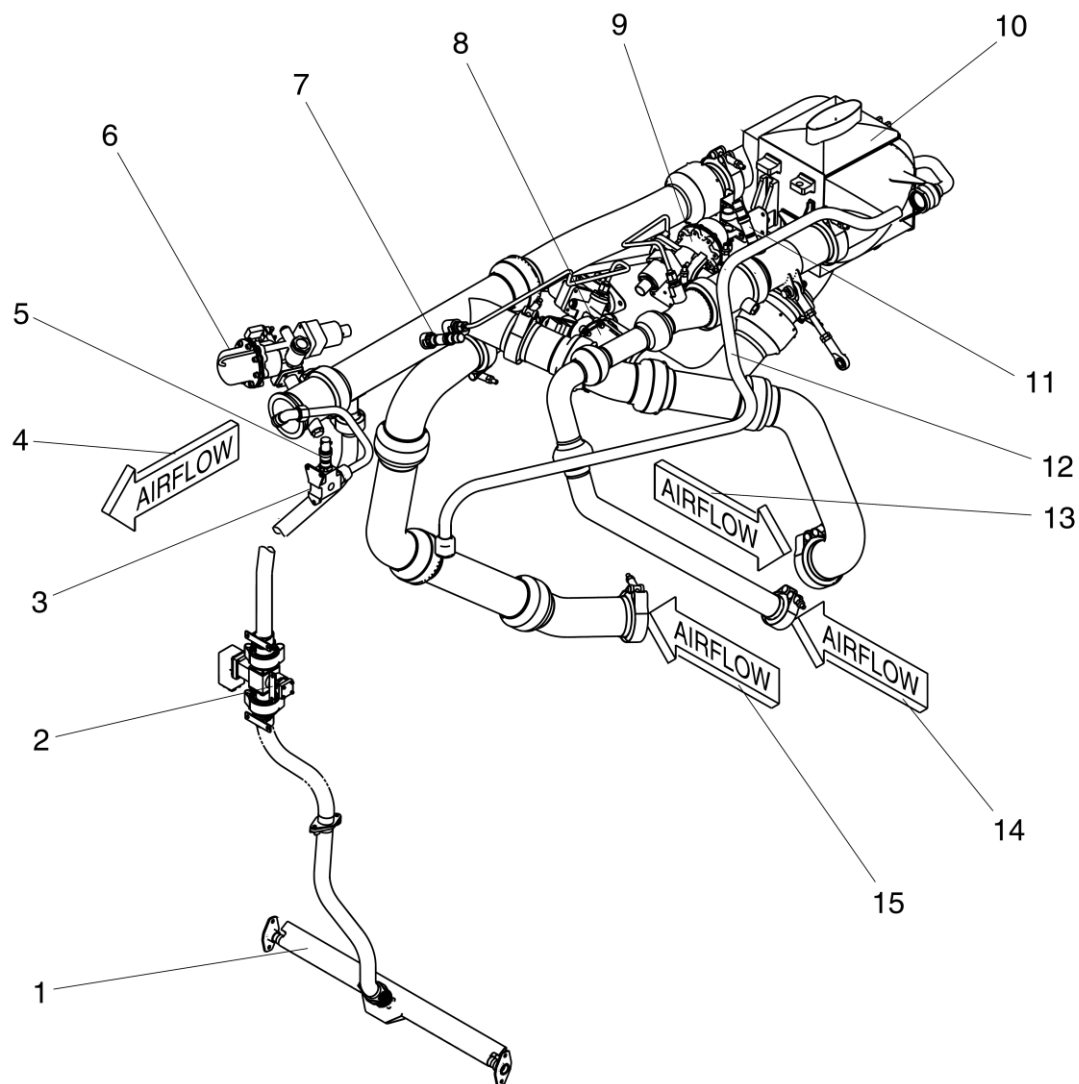
A smaller deice precooler, attached to the high stage precooler, further reduces the temperature of the bleed air sent to the deice system. This precooler uses a continuous flow (no flow control valves) of P2.2 bleed air and operates during selection of either P3.0 or P2.7 bleed air.

The flow of P2.2 bleed air through the deice precooler absorbs heat from the P3.0 or P2.7 bleed air. This reduces the temperature of the P3.0 or P2.7 bleed air to acceptable levels before it flows to the deice system.

When the deice system is in operation, and the P3.0 bleed air pressure decreases below 77 psi. (as detected by the bleed stage pressure switch), the HPSOV opens. This function is independent of the ECU.

A discrete signal from the deice system and an open discrete signal from the HPSOV tell the ECU that P3.0 air is selected for deice use. The ECU adjusts bleed air flow as necessary. The ECU uses a differential pressure (flow) sensor to measure the bleed air flow through each nacelle shutoff valve.

The ECU uses this bleed airflow information to balance the flow of bleed air from both engines to the ECS. Each digital channel in the ECU modulates its nacelle shutoff valve to maintain 50% total bleed air flow from its engine. Bleed sharing can occur only when both engines are in operation and both digital channels are functional. There is no analog channel function for bleed air sharing.



LEGEND

1. Oil Cooler Ejector.
2. Oil Cooler Ejector Shutoff Valve.
3. Leak Detection Housing.
4. Bleed Air Supply.
5. Duct Leak Overtemperature Switch.
6. Nacelle Shutoff Valve.
7. Overtemperature Switch.
8. P2.2 Shutoff Valve.
9. (P3.0) High Pressure Shutoff Valve (HPSOV).
10. Precooler.
11. Precooler Temperature Switch.
12. Deice Supply Line.
13. P 2.7 Air.
14. P 3.0 Air.
15. P 2.2 Air.

Figure 36-6 Bleed Air System - Nacelle

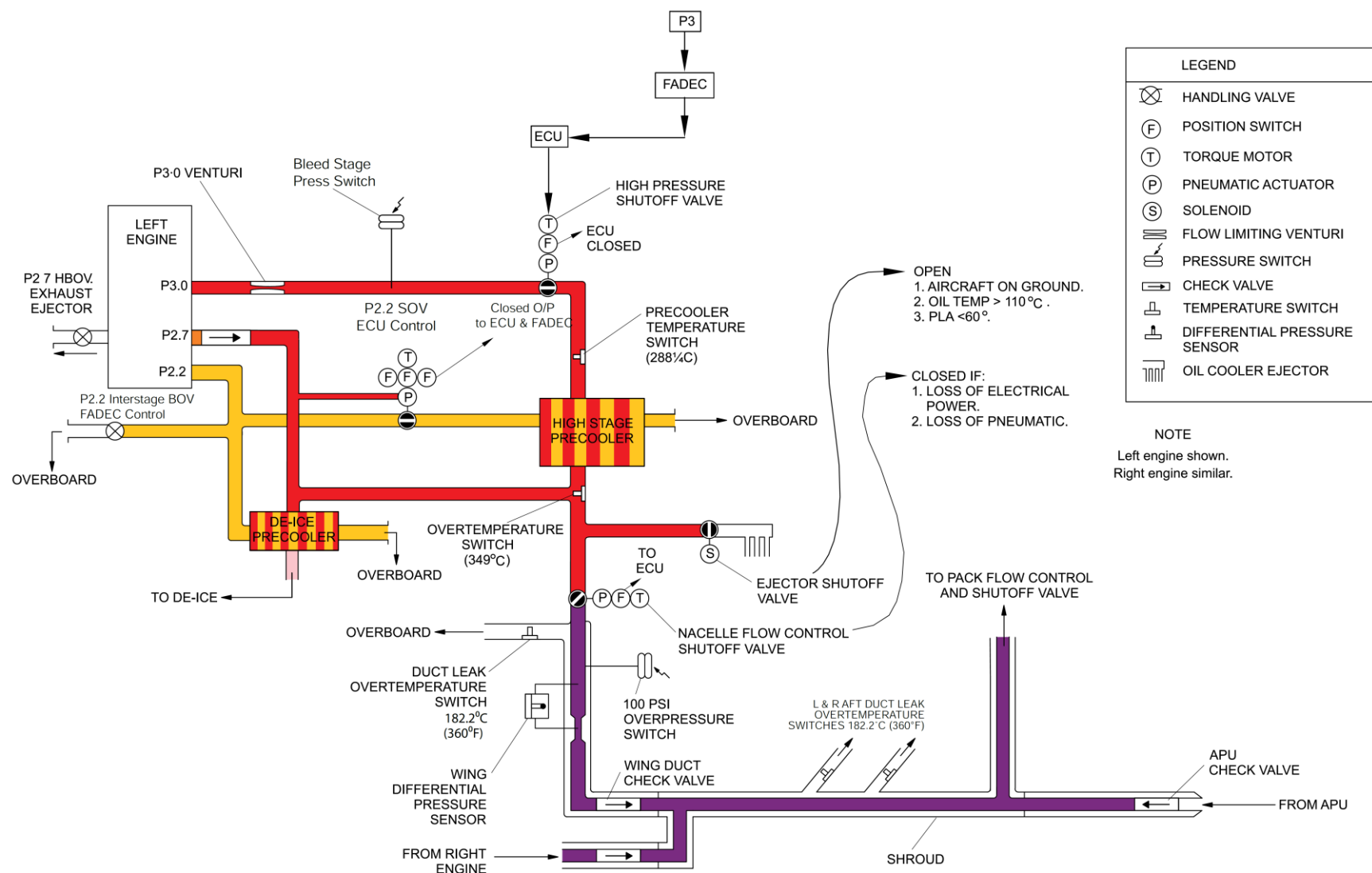


Figure 36-7 Bleed Air System Schematic – P3.0 Open

ECU Operation

Refer Figure 36-8

The ECU is the interface between the pilot and crew control panels and the mechanical and electrical components of the ECS (which includes the Bleed Air System). The ECU uses sensors and switches to monitor temperature, pressure, flow rates and valve positions. The ECU responds to the pilot and crew commands to configure the Bleed Air System for bleed source selection. The ECU and the Bleed Air System have redundant configurations to permit continued ECS operation with mechanical and electrical component malfunctions.

The ECU is divided into a left digital channel with a left backup analog channel and a right digital channel with a right backup analog channel. The left digital channel and right analog channel receive electrical power from the 28 Vdc L MAIN bus. The right digital channel and left analog channel receive electrical power from the 28 Vdc R MAIN bus. This permits continued left and right channel control if one 28 Vdc bus loses power. For example, if the left digital channel loses electrical power because of bus failure, the left analog channel is still powered by the remaining bus.

Inside the ECU, the digital channels use an RS 422 digital data bus to communicate with each other. The ECU uses the ARINC 429 digital data bus to communicate to other aircraft systems. The ARINC 429 receivers of both digital channels are connected to the aircraft wiring. Only the ARINC 429 transmitter of the left digital channel is connected to the aircraft wiring.

The left and right digital channels share control of the nacelle shutoff valves. At the start of a flight, the ECU selects one digital channel to control the shared components of the bleed air system. The digital channel in control is responsible for all functions applicable to the shared components. Loss of the digital channel in control causes the other digital channel to take control of the shared components.

The left digital channel and left analog channel share control of the left nacelle shutoff valve. If the left digital channel loses power or fails, the left analog

channel automatically takes control of this valve. The right digital channel and right analog channel share control of the right nacelle shutoff valve. If the right digital channel loses power or fails, the right analog channel automatically takes control of this valve.

If the pilot selects the manual mode, the left digital channel (responsible for the auto mode) transfers control of the left nacelle shutoff valve to the left analog channel. This description also applies to the right digital channel and right analog channel which control the right nacelle shutoff valve.

The analog channels function as backup controllers for the digital channels. One or both analog channels take control of the functions only when one or both digital channels lose power or malfunction. If the PACKS switch on the AIR CONDITIONING control panel is set to MAN (manual), the analog channels are in control of the ECS (including the BAS). The analog channels keep the nacelle shutoff valves fully open (50 mA drive current to the torque motors), but defaults the HPSOVs closed. This disables the high stage (P3.0) bleed air and selects the low stage (P2.7) bleed air. The protection circuits remain active but balanced bleed air (bleed sharing) from both engines is not possible. This is because the analog channels do not have inputs (differential pressure sensors) to sense bleed air flow.

The analog channels keep the nacelle shutoff valves fully open during these conditions:

- The BLEED 1 and/or BLEED 2 switches on the AIR CONDITIONING control panel turned on
- No overtemperature, overpressure or bleed leak indications are active.

The ECU (digital and analog channels) examines the state of all three switches: overtemperature switch (On aircraft with ModSum 4Q113658 or SB84-36-03 incorporated, the overtemperature switch is replaced with the overtemperature sensor), overpressure switch, and four duct leak switches. Each digital channel examines the state of the protective switches for the BAS on its side of the aircraft. A discrete signal from one or more switches activates a protective bleed shutoff (closes nacelle shutoff valve, HPSOV, and

P2.2 shutoff valve). Built-In-Test (BIT) logic determines which switch caused the BAS shutdown.

The digital channel will keep its side latched off until the switch that tripped has reset and the BLEED 1 or BLEED 2 switch is set to off. The applicable analog channel supplies backup BAS protection if a digital channel loses power or fails. Bleed air (left or right) remains shut off until the pilot sets the BLEED 1 or BLEED 2 switch to OFF and then turns it on again.

Each digital and analog channel generates a discrete signal to turn on a BLEED HOT caution light which signals a failure of the BAS. If both the digital and analog channel on one side fail, the related BLEED HOT caution light will not turn on.

The ECU does high level operational checks of the ECS system as part of continuous BIT. For the ECS, the tests examine that the pack flow is within permitted limits and that the pack discharge temperature is reasonable. Failure of these high level tests causes the ECU to do low level tests to identify the failed component. The ECU also performs self tests on its digital and analog channels and the applicable interfaces.

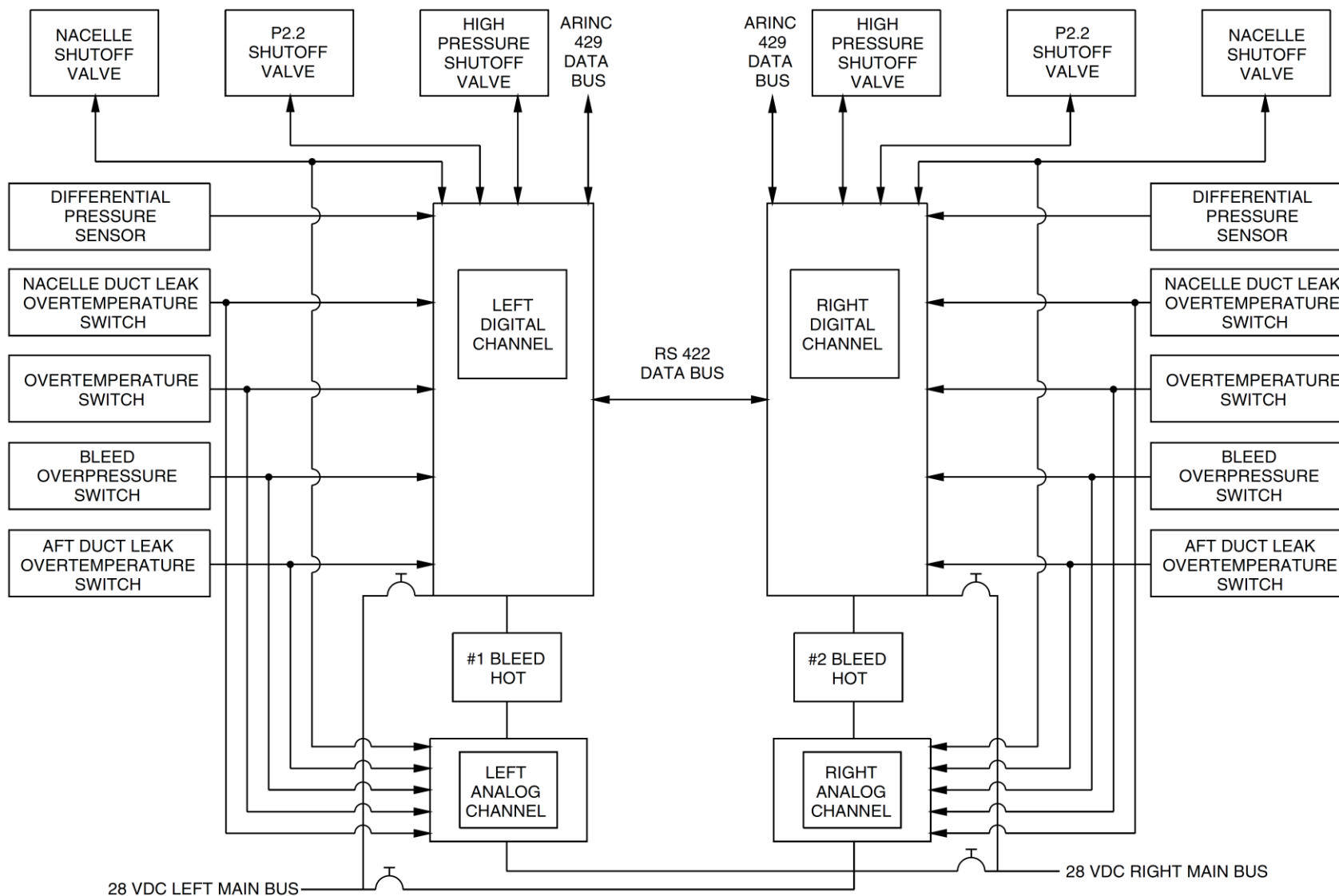
The high pressure shutoff valves, nacelle shutoff valves and the APU shutoff valve each have a position switch to identify when the valve is closed. The P2.2 SOV has three electrically isolated position switches to identify an open or closed valve position.

The digital channels use one of the three switches for its Built-In- Test (BIT). During its BIT, the ECU compares the commanded state (open or closed) of the valve to the actual state identified by the position switch. BIT logic can also examine the signals from the pressure and temperature sensors to determine if they function correctly.

When the ECU detects a BIT fault, it saves an applicable fault code in its non-volatile fault memory. A failure detected in a digital channel causes the ECU to transmit the fault code signal to the Central Diagnostic System (CDS).

During ground operation, the digital channels of the ECU respond to the INITIATE TEST and FAULT RESET input signals from the CDS as follows:

- The INITIATE TEST signal causes the ECU to report (transmit) BIT faults via the ARINC-429 digital data bus
- The FAULT RESET signal causes the ECU to erase all BIT faults saved in non-volatile memory which are transmitted via ARINC-429.



NOTE: ALL CHANNELS ARE LOCATED IN ECU CONTROLLER

Figure 36-8 Bleed Air Control Block Diagram

Bleed Air System - Component Description

P3.0 Ducts

[Refer Figure 36-9](#)

High pressure ducting, installed between the engine (P3.0) bleed air port and the pre-cooler inlet duct, supplies air flow at low power settings.

The high-pressure ducting in the nacelle is enclosed in a composite shroud from the bleed air port to the high pressure shut off valve. This reduces the possibility of ignition in the event of a fuel leak.

P3.0 Venturi

[Refer Figure 36-9](#)

The P3 bleed air adapter (P3.0 Venturi) installed on the engine gas generator case limits the amount of hot, high pressure air entering the bleed air system.

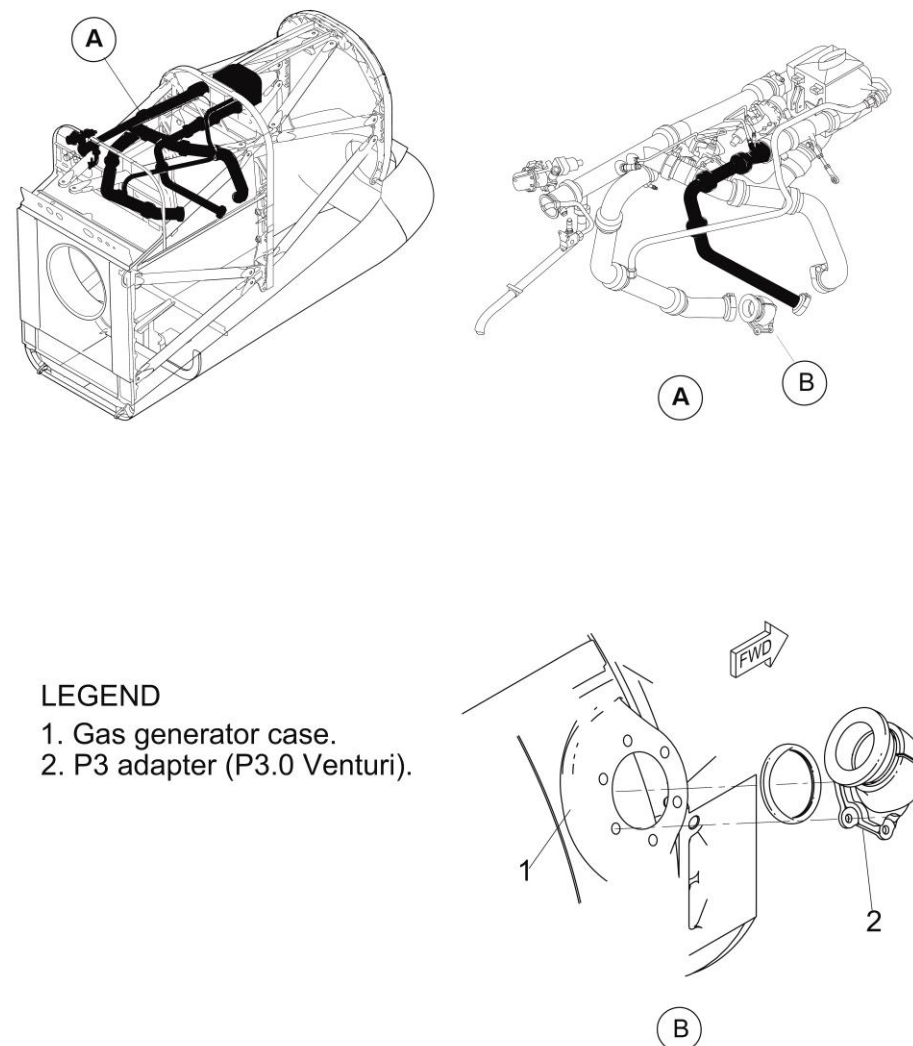


Figure 36-9 P3.0 Ducts and P3.0 Venturi – Location

Bleed Stage Pressure Switch

[Refer Figure 36-10](#)

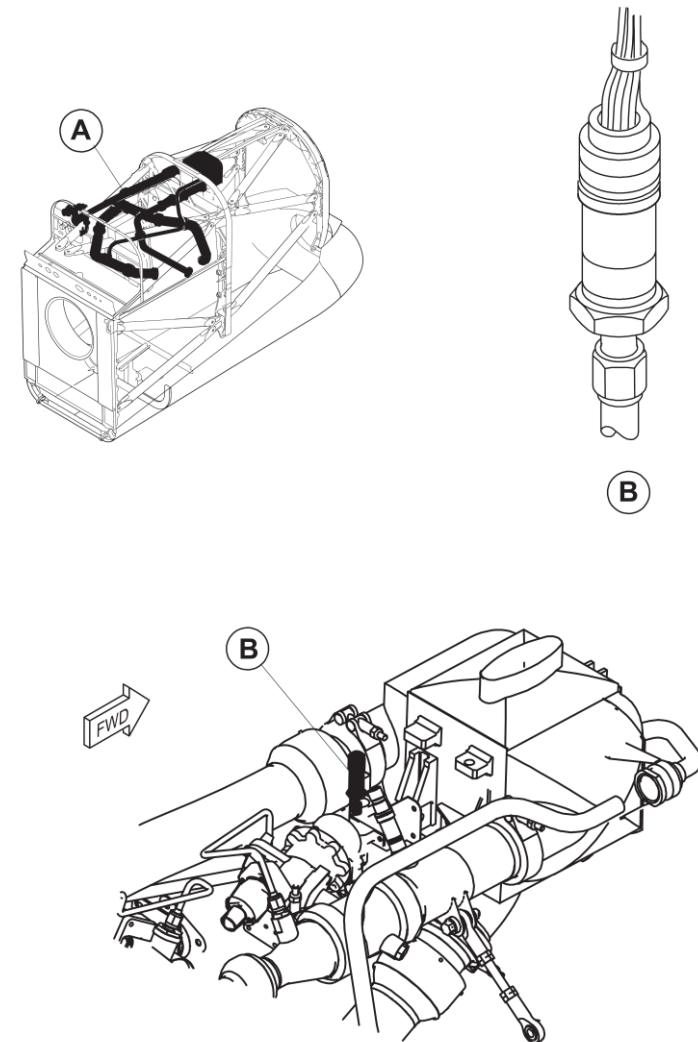
The bleed stage pressure switch is installed in a bracket aft of the precooler. The switch controls the HPSOV independently of the ECU for deice operation.

When the bleed switches are selected off or the deice system is selected on, the HPSOV is controlled by the bleed stage pressure switch.

The switch:

- Opens at 82 psi (565 kPa) increasing pressure
- Closes at 77psi (531 kPa) decreasing pressure

A discrete signal from the deice system and an open discrete signal from the HPSOV identifies to the ECU that high stage (P3.0) bleed air is selected for deice use. The ECU adjusts bleed air flow as required.



A BLEED AIR SYSTEM

Figure 36-10 Bleed Stage Pressure Switch – Location

High Pressure Shutoff Valve

[Refer Figure 36-11](#)

The high pressure shutoff valve (HPSOV) is a pneumatically operated, torque motor actuated butterfly valve. The HPSOV is installed in line with the high pressure ducting between the engine (P3.0) bleed air port and the precooler.

The high pressure ducting in the nacelle is enclosed in a composite shroud from the bleed port to the HPSOV. This reduces the possibility of ignition in the event of a fuel spillage. The valve has a wire mesh filter upstream of the torque motor to protect the servo from contamination.

The HPSOV opens when the torque motor is energized. When the HPSOV is in the open position, it allows P3.0 air to flow through to the precooler to supply the ECS and/or the deice systems.

The ECU uses high pressure (P3.0) information supplied by FADEC to control the opening and closing of the HPSOV. The ECU opens the HPSOV at low engine power settings.

The HPSOV has a closed position switch which informs the ECU of the status of the valve. The bleed stage pressure switch can open the HPSOV independently from the ECU to supply bleed air to the deice system.

Bleed Switch	De-Ice Switch	HPSOV Controlled By:
OFF	OFF	Bleed Stage Pressure Switch
OFF	ON	Bleed Stage Pressure Switch
ON	ON	Bleed Stage Pressure Switch
ON	OFF	ECS ECU

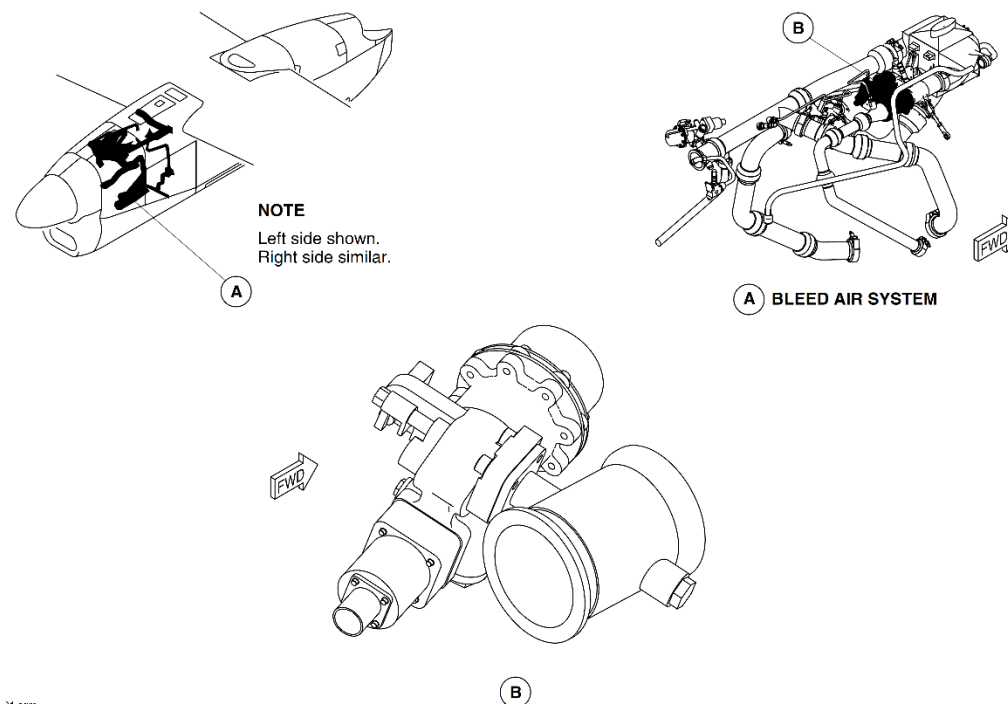


Figure 36-11 High Pressure Shutoff Valve – Location

Precooler Temperature Switch

[Refer Figure 36-12](#)

The precooler temperature switch is installed in the ducting between the HPSOV and the precooler. The precooler switch monitors the temperature of the engine bleed air entering the precooler.

If the bleed air entering the precooler is greater than 550°F (288 °C), the switch will open sending a signal to the ECU to open the P2.2 shutoff valve.

The ECU will use this signal to command the P2.2 valve open allowing cooling air to pass through the pre-cooler.

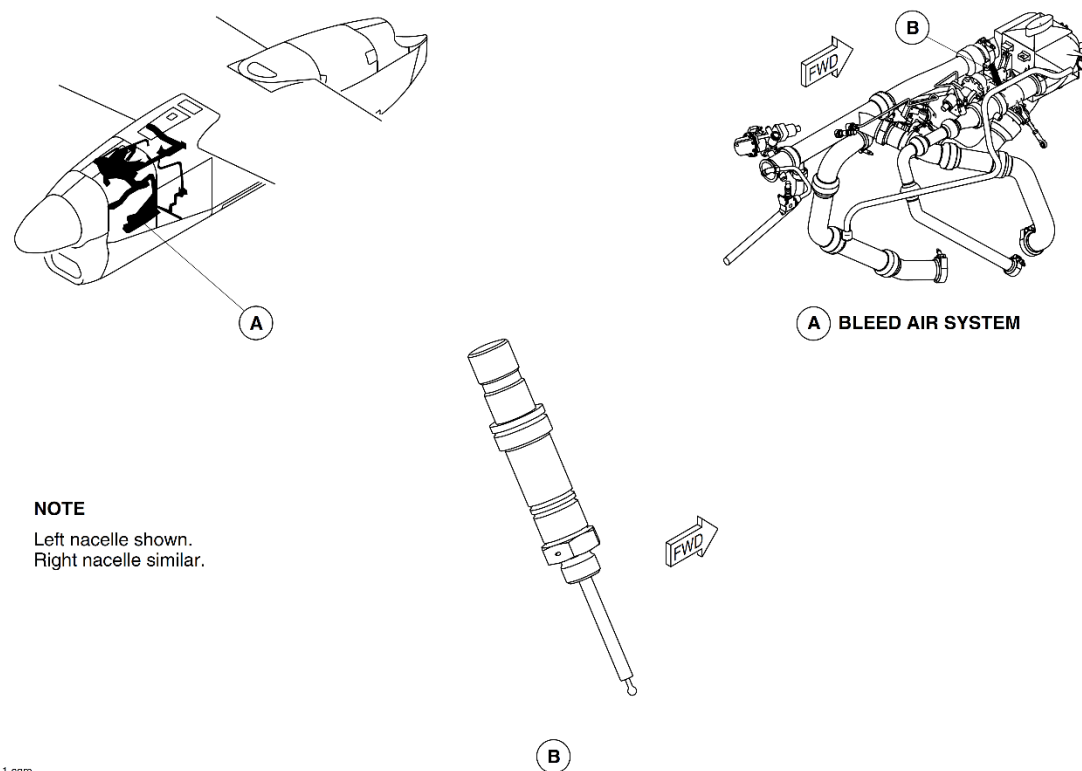


Figure 36-12 Precooler Temperature Switch – Location

Precooler

[Refer Figure 36-13](#)

The precooler is installed near the top of the engine. It is a two-part plate and fin, cross flow, air to air heat exchanger which uses low temperature (P2.2) bleed air to cool high pressure bleed air.

The main heat exchanger cools the temperature of the high pressure (P3.0) bleed air before it is supplied to the ECS. The small heat exchanger cools either the high pressure (P3.0) or the low pressure (P2.7) bleed air before it is supplied to the deice system.

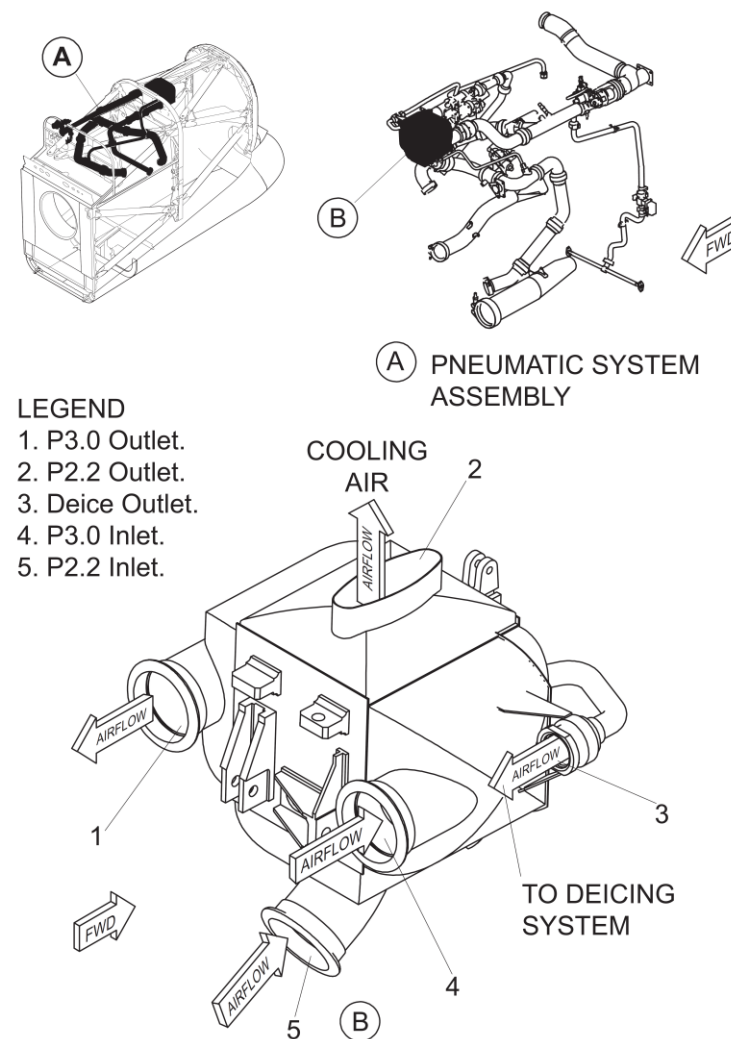


Figure 36-13 Precooler Assembly - Location

Overtemperature Switch/Sensor

[Refer Figure 36-14](#)

The overtemperature switch is installed in the precooler outlet duct. It is a protective switch that monitors nacelle bleed air temperature. If the temperature of the bleed air entering the nacelle ducts is greater than 660°F (349 °C), the overtemperature switch will open. This sends a signal to the ECU to close the nacelle shutoff valve for the affected side. The ECU will then turn on the appropriate BLEED HOT caution light.

On aircraft with ModSum 4Q113658 or SB84-36-03 incorporated, the overtemperature switch is replaced with a sensor. This sensor is a Resistance Temperature Device (RTD). It gives a resistance, which is proportional to the temperature it senses.

If the temperature of the bleed air entering the nacelle ducts is greater than 660°F (349 °C), the overtemperature sensor sends a signal to the ECU to close the nacelle shutoff valve for the affected side. The ECU will then turn on the appropriate BLEED HOT caution light.

The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- NSOV
- HPSOV
- P2.2 valve.

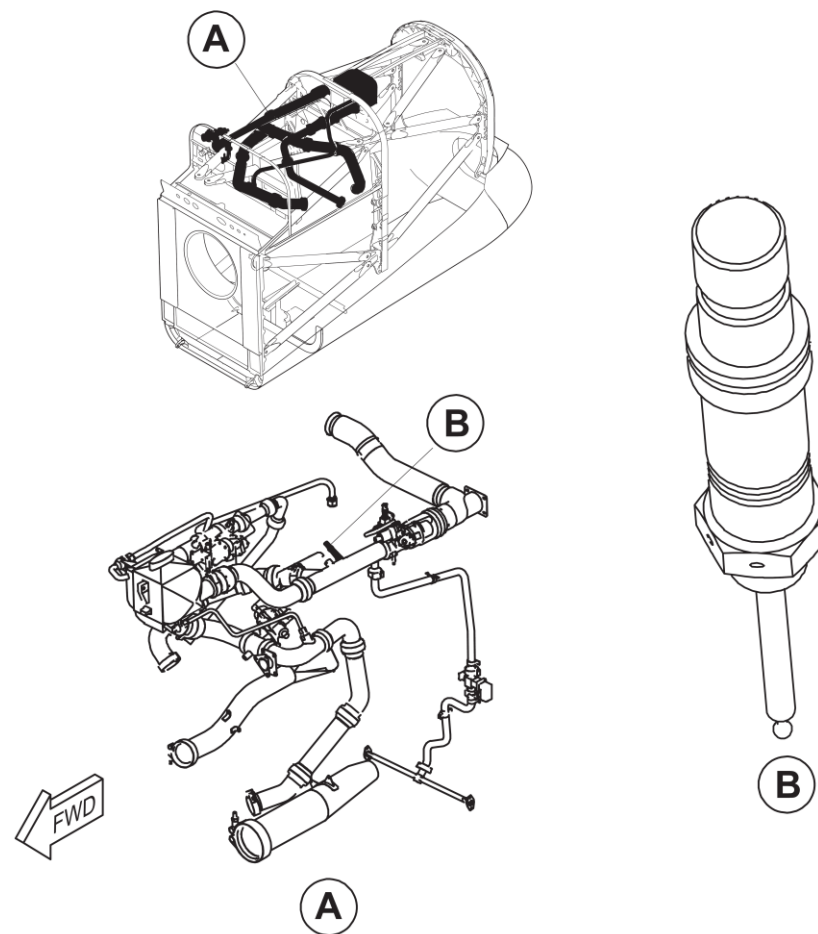


Figure 36-14 Overtemperature Switch/Sensor – Location

Nacelle Shutoff Valve

[Refer Figure 36-15](#)

The nacelle shutoff valve is a pneumatically operated, torque motor actuated butterfly valve. The nacelle shutoff valve is installed in line with the precooler outlet duct and the wing ducting. The valve has a wire mesh filter upstream of the torque motor to protect the servo from contamination.

Selection of the BLEED 1 and/or BLEED 2 switches on the AIR CONDITIONING control panel opens the related nacelle shutoff valve. The ECU uses the nacelle shutoff valve to control the flow of bleed air to the ECS.

The ECU regulates the nacelle shutoff valve according to the settings of the BLEED, PACKS and MIN/NORM/MAX switches on the AIR CONDITIONING control panel.

The ECU also regulates the nacelle shutoff valve in each nacelle to supply 50% of the total flow demand. This ensures equal flow sharing between each main engine.

The nacelle shutoff valve has a closed position switch which informs the ECU of the status of the valve.

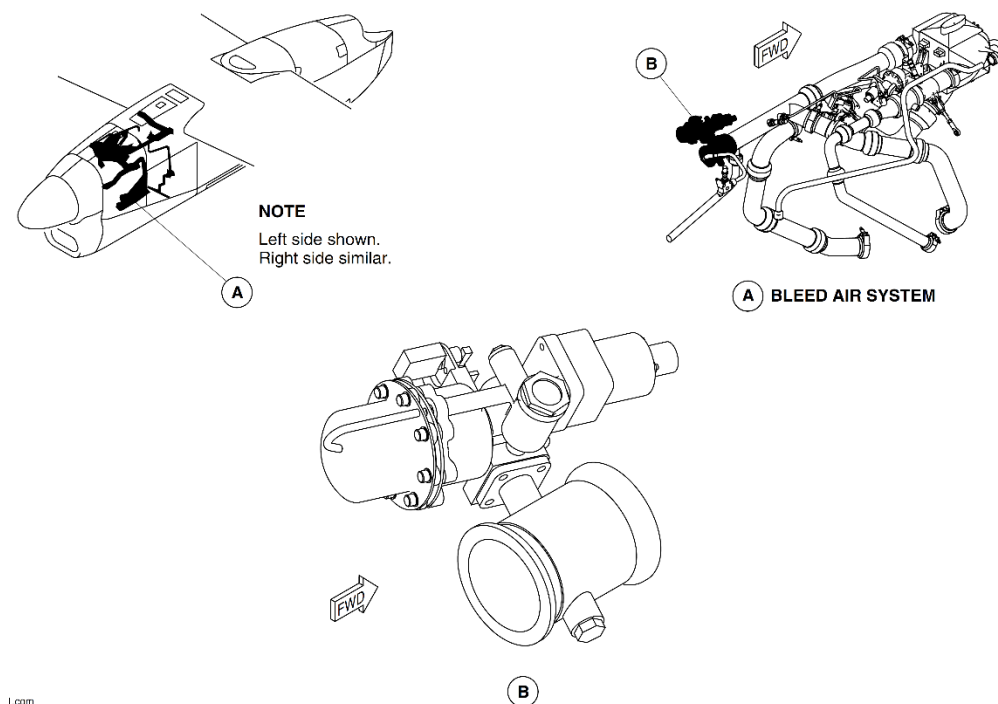


Figure 36-15 Nacelle Shutoff Valve – Location

Nacelle Duct Leak Overtemperature Switch

[Refer Figure 36-16](#) & [Refer Figure 36-17](#)

The nacelle duct leak overtemperature switch is installed on a bracket next to the nacelle shutoff valve. It detects a bleed air leak in the nacelle ducting.

If the temperature in the nacelle ducting is greater than 360 °F (182.2 °C), the switch opens sending a signal to the ECU to close the appropriate nacelle SOV.

The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- NSOV
- HPSOV
- P2.2 SOV

The ECU also sends a fault code to the Central Diagnostic System (CDS) identifying a duct leak will bring on the related BLEED HOT caution light.

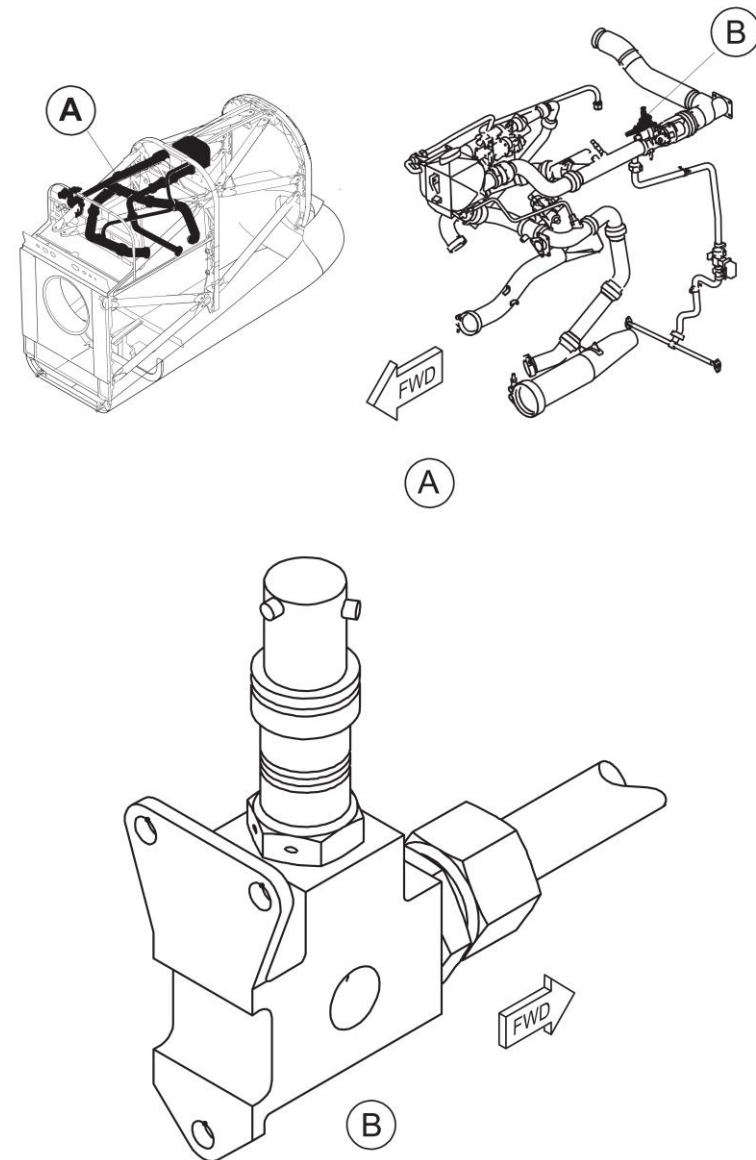


Figure 36-16 Nacelle Duck Leak Overtemperature switch – Location

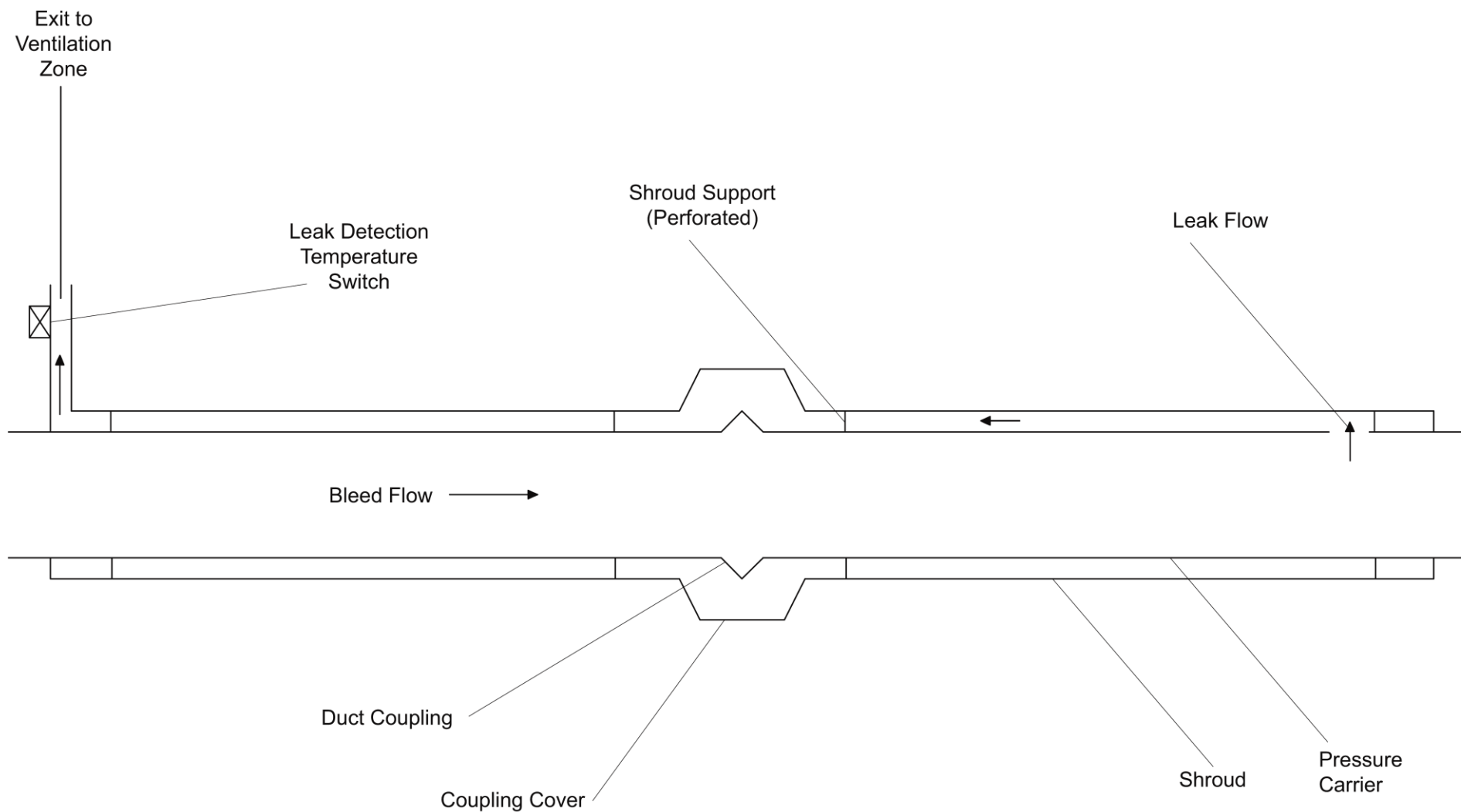


Figure 36-17 Nacelle Duck Leak Detection – Concept Schematic

P2.2 Shutoff Valve

[Refer Figure 36-18](#)

The P2.2 shutoff valve is a pneumatically operated, torque motor actuated butterfly valve. The valve has a wire mesh filter upstream of the torque motor to protect the servo from contamination.

The P2.2 shutoff valve is installed in line with the P2.2 low pressure ducting and the P2.2 inlet at the bottom of the precooler. It opens to supply cooling air through the precooler.

The precooler temperature switch controls its operation. The valve has two closed position switches to supply valve position information to the ECU and to FADEC.

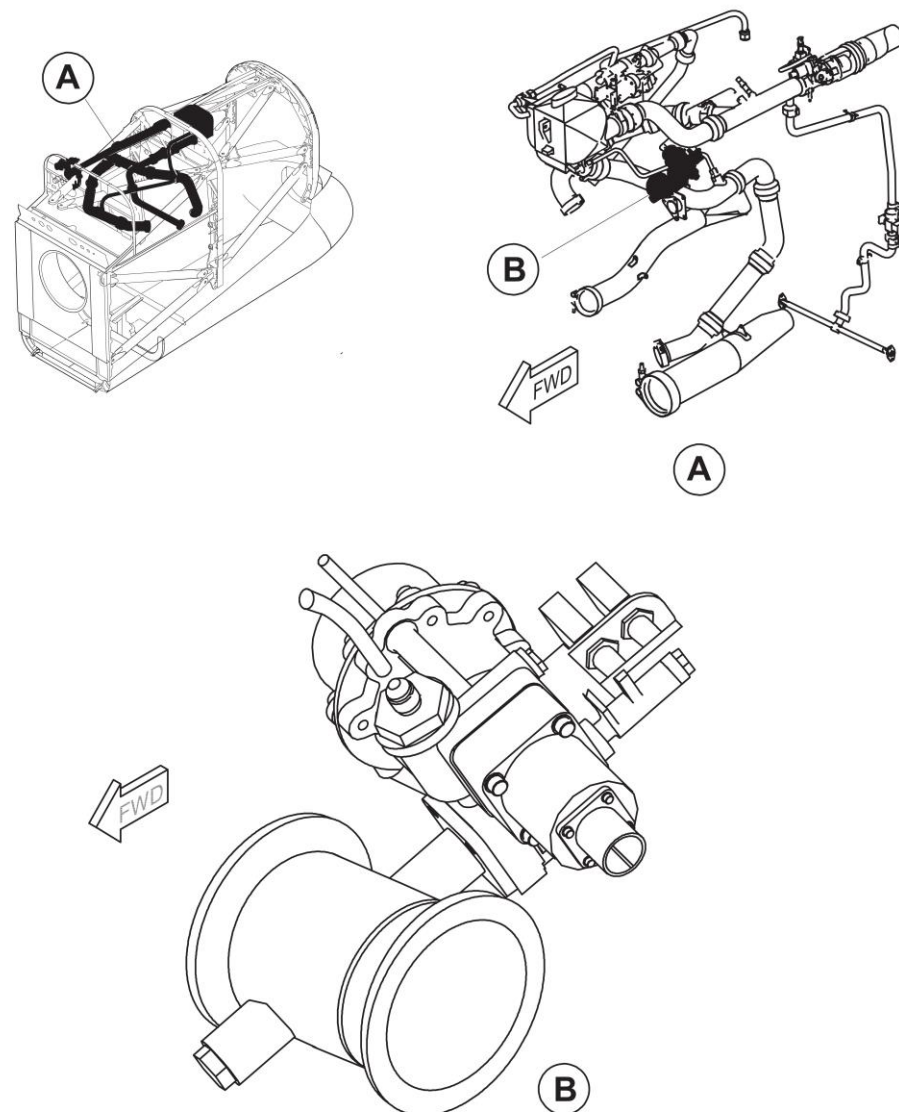


Figure 36-18 P2.2 Shutoff Valve - Location

Wing Bleed Air Ducts

[Refer Figure 36-19](#)

Titanium wing bleed air ducts are installed on the wing front spar and in the wing box on top of the fuselage in the centre section of the wing.

The wing bleed air ducts are an assembly of the following pieces of bleed air ducts:

- Firebox duct
- Inboard firebox duct
- Wing spar duct
- Fairing duct
- Wingbox duct
- Wye duct.

The ducts contain wing differential pressure sensors and wing duct check valve overpressure switches.

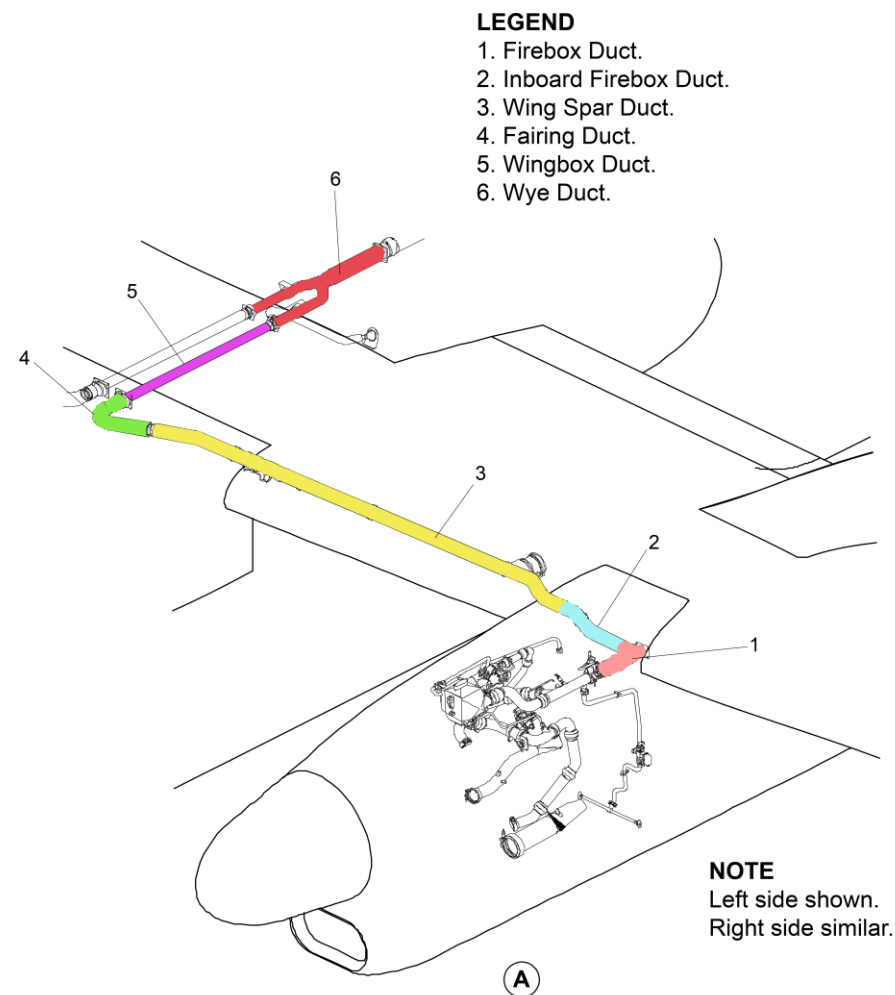


Figure 36-19 Wing Bleed Air Ducts

Overpressure Switch

[Refer Figure 36-20](#)

The overpressure switch is installed on a bracket in the wing leading edge and is connected to the bleed air ducting with rigid tubing. It is a protective switch that monitors nacelle bleed pressure.

If the pressure in the nacelle ducting is greater than 100 psi (690 kPa), the overpressure switch will close.

The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- NSOV
- HPSOV
- P2.2 SOV

The ECU also sends a signal to turn on the appropriate BLEED HOT caution light and log a fault code in the Central Diagnostics System (CDS).

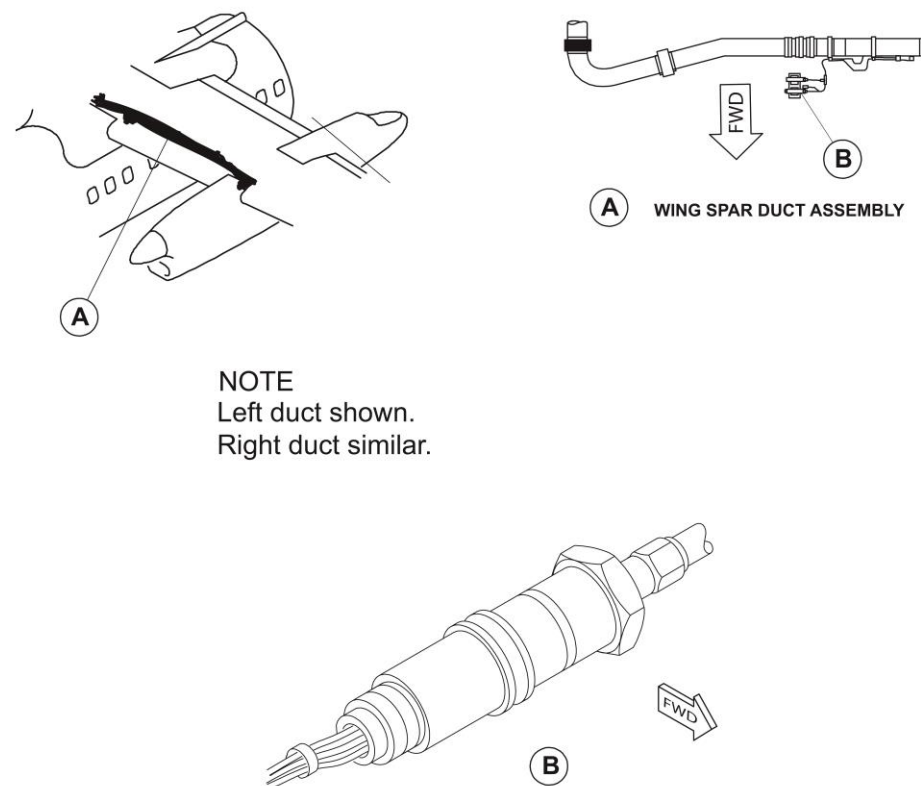


Figure 36-20 Overpressure Switch – Location

Wing Differential Pressure Sensor

[Refer Figure 36-21](#)

The wing differential pressure sensor is installed on a bracket in the wing leading edge and is connected to bleed air ducting with rigid tubing. The differential pressure sensor has two pressure ports.

One port senses the pressure upstream of the wing venturi and the other port senses the pressure at the throat of the venturi. The differential pressure data is sent to the ECU.

The ECU uses this data together with pressure and temperature data to regulate the bleed air flow through the nacelle shutoff valve.

Wing Duct Check Valve

[Refer Figure 36-21](#)

The wing duct check valve is a dual flapper, spring loaded check valve. The valve is installed in the bleed air ducting along the wing leading edge. It is held open by bleed air from the engine.

The check valve closes during single engine operation to prevent bleed air from the operating engine flowing into the bleed air ducting of the opposite engine.

The check valve also prevents reverse flow into both engine bleed air systems when the APU is supplying the bleed air.

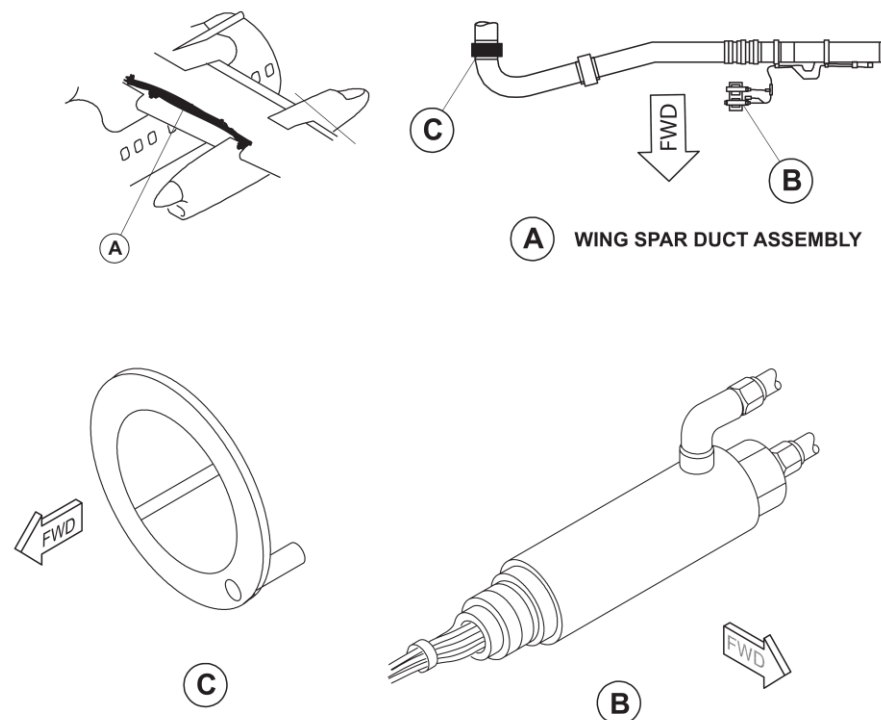


Figure 36-21 Wing Differential Pressure Sensor and Check Valve

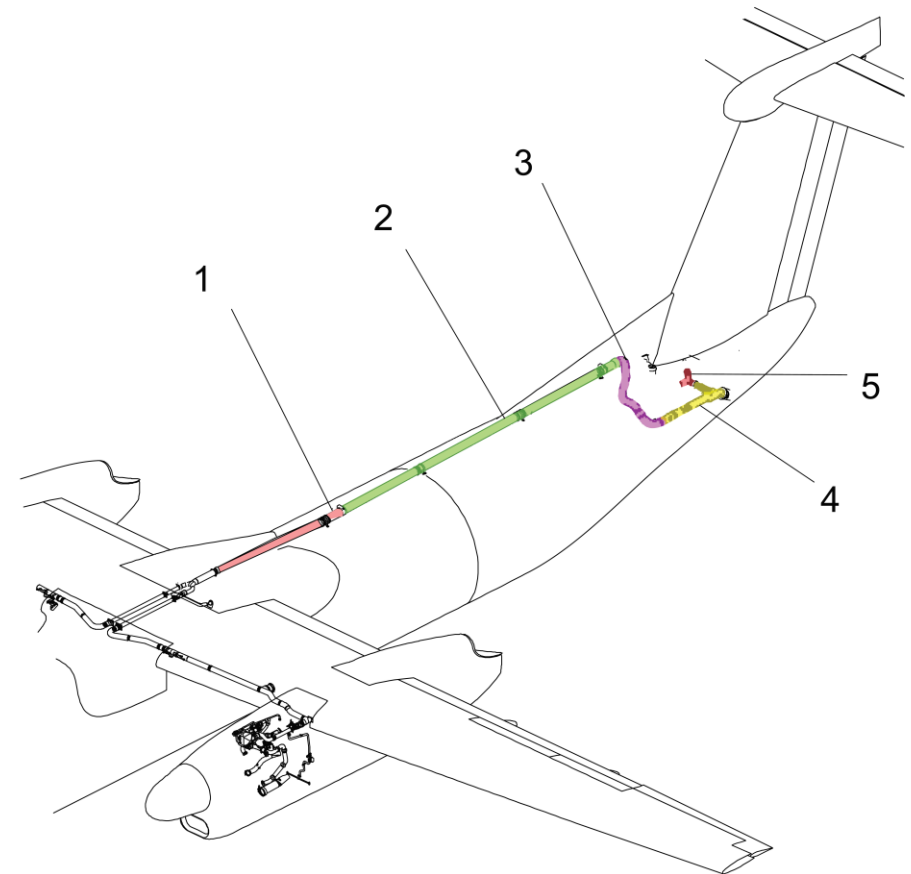
Dorsal and Aft Fuselage Bleed Air Ducts

[Refer Figure 36-22](#)

Titanium dorsal bleed air ducts, installed on top of the fuselage, run from the wye duct in the wing center section along the top of the fuselage to the dorsal fin at the aft fuselage.

The dorsal ducting incorporates a pair of bellows ball joints that add flexibility to the assembly.

The dorsal and aft fuselage bleed air ducts have shrouds and contain a duct leak over-temperature switch and APU duct leak overtemperature switch.



LEGEND

- | | |
|-------------------------|-------------------------------|
| 1. Forward Dorsal Duct. | 3. Aft Fuselage Duct. |
| 2. Aft Dorsal Duct. | 4. Aft Fuselage Tee Duct. |
| | 5. Heat Exchanger Inlet Duct. |

Figure 36-22 Dorsal and Aft Fuselage Bleed Air Ducts

APU Differential Pressure Sensor

[Refer Figure 36-23](#)

The APU differential pressure sensor is installed on a bracket on the APU bleed air ducting in the aft fuselage. It is connected to the bleed air ducting by rigid tubing.

The differential pressure sensor has two pressure ports. One port senses the pressure upstream of the APU venturi and the other port senses the pressure at the throat of the venturi. The differential pressure data is sent to the ECU.

The ECU uses this data together with pressure and temperature data to regulate the bleed air flow through the pack flow control and shutoff valve.

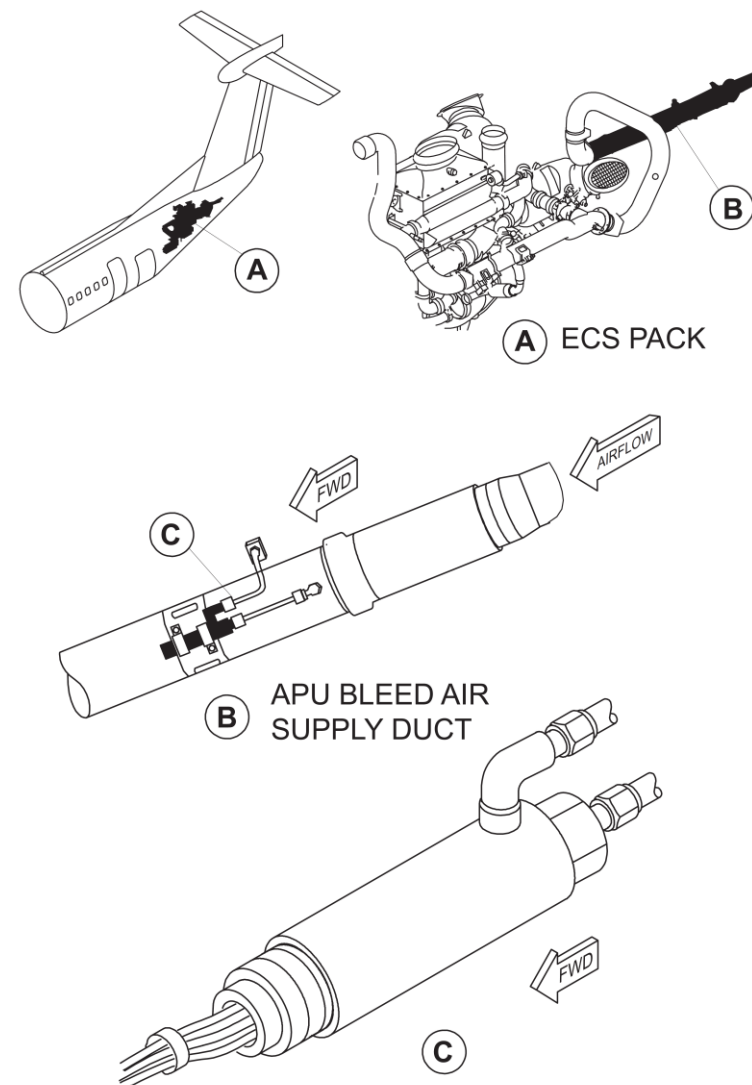


Figure 36-23 APU Differential Pressure Sensor - Location

Aft Fuselage Duct Leak Overtemperature Switches

[Refer Figure 36-24](#)

Two aft fuselage duct leak overtemperature switches are installed on a bracket on the aft fuselage tee duct. They detect a bleed air leak in the engine or APU bleed air ducting near the air conditioning pack inlet.

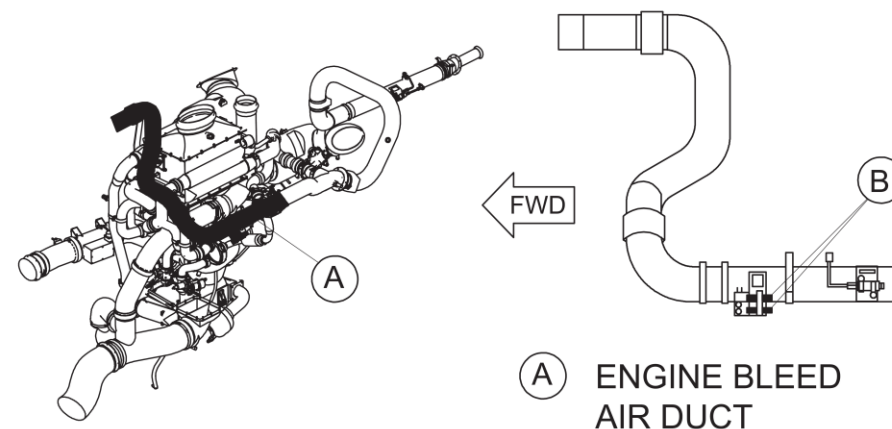
If the temperature in the ducting is greater than 360 °F (182.2 °C), the switch contacts will open.

The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- NSOV
- HPSOV
- P2.2 SOV

The APU FADEC will close the APU bleed valve. The BL AIR OPEN advisory light on the APU panel goes OFF.

The ECU also illuminates the two BLEED HOT caution lights and records a fault code in the CDS to show a duct leak.



NOTE
Upper component shown.
Lower component similar.

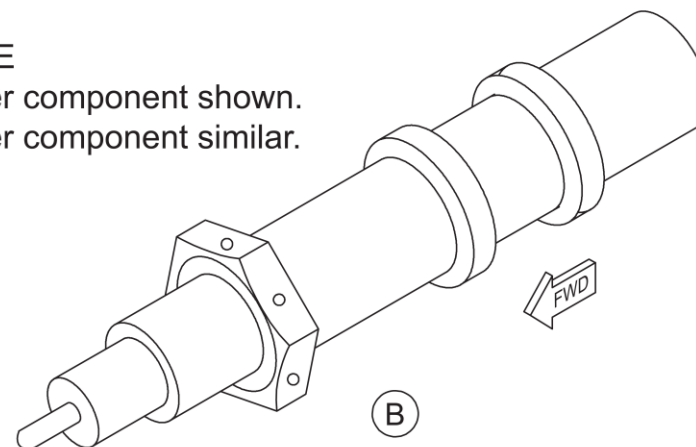


Figure 36-24 Aft Fuselage Duct Leak Overtemperature Switches

APU Bleed Air Ducts

[Refer Figure 36-25](#)

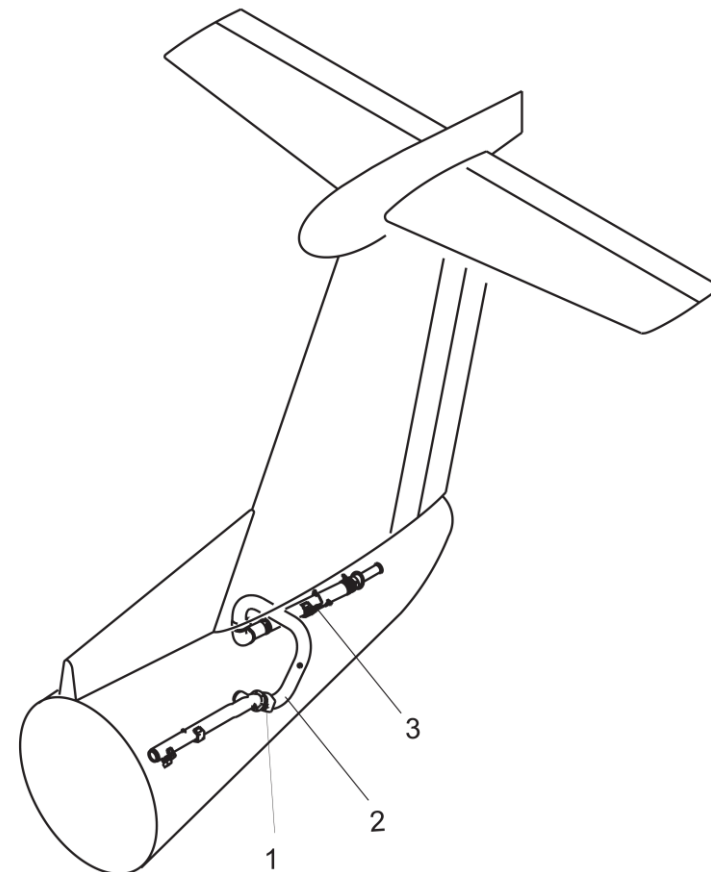
Shrouded titanium APU bleed air ducts, installed aft of the rear pressure bulkhead, run forward through the APU firewall and join the aft fuselage duct upstream of the PFCSOV.

APU Check Valve

[Refer Figure 36-25](#)

The APU check valve is a dual flapper, spring loaded check valve. The valve is installed in line with the APU bleed air ducting in the aft fuselage.

It is held open by bleed air coming from the APU. It allows bleed air to flow to the air conditioning pack but prevents reverse flow to the APU from the engine bleed air system.



LEGEND

- 2. APU Duct Check Valve.
- 2. APU Mid Duct.
- 3. APU Forward Duct.

Figure 36-25 APU Bleed Air Ducts and Check Valve

Environmental Control System Electronic Control Unit (ECSECU)

Refer Figure 36-26

The ECU, installed under the center floor boards, responds to pilot switch and sensor commands to configure the BAS sources. The ECU and the BAS have redundant configurations to permit continued ECS operation with mechanical and electrical component malfunctions.

The ECU has two channels designated left and right. Each channel has a digital channel for automatic control and an analog channel for manual control. The analog channel also provides limited backup control should the digital channel fail.

The ECU controls and monitors the bleed air system for the following indications:

- Bleed flow
- Bleed pressure
- Bleed temperature
- Engine bleed supply
- Bleed duct leak detection
- Valve positions.

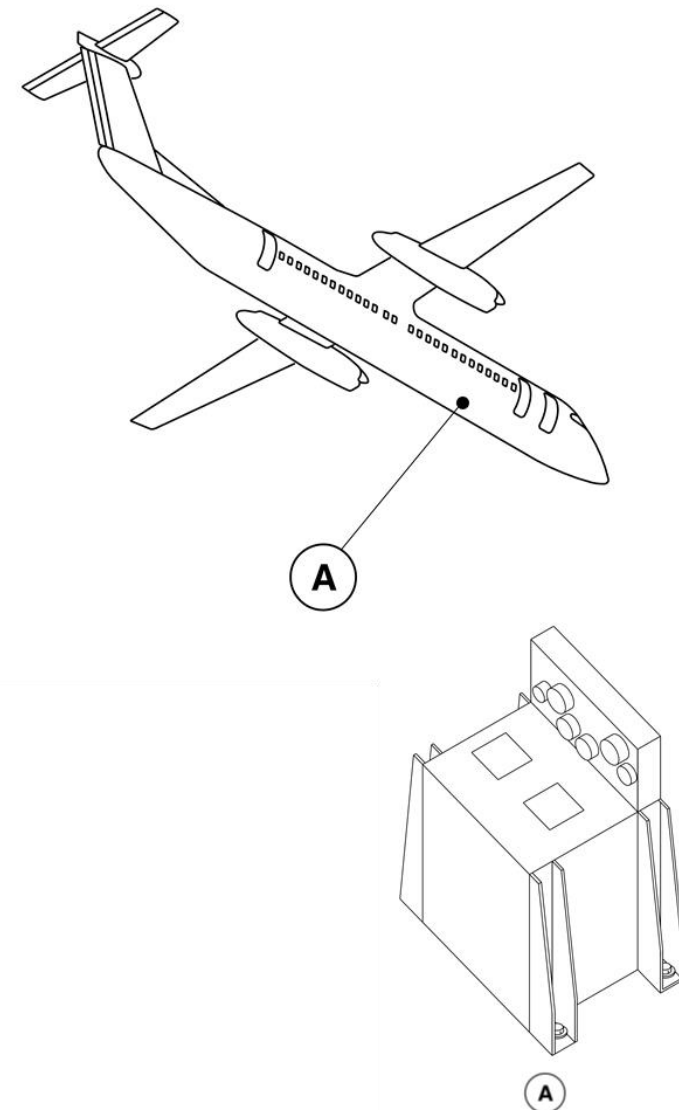


Figure 36-26 ECSECU - Location

Controls and Indications

Bleed Control Switches

[Refer Figure 36-27](#)

The bleed air switches are installed in the flight compartment on the air conditioning control panel.

BLEED 1 and BLEED 2 switches: Selection to ON signals the ECU to open and control NSOV.

Bleed Control Selector

[Refer Figure 36-27](#)

The BLEED control selector is a three-position rotary selector that is used to control the amount of bleed air flow from the two engines.

MIN: Causes the ECU to modulate the nacelle flow control shut-off valve to minimum air flow.

NORM: Causes the ECU to modulate the nacelle flow control shutoff valve to normal air flow.

MAX: Causes the ECU to modulate the nacelle flow control shut-off valve to maximum air flow.

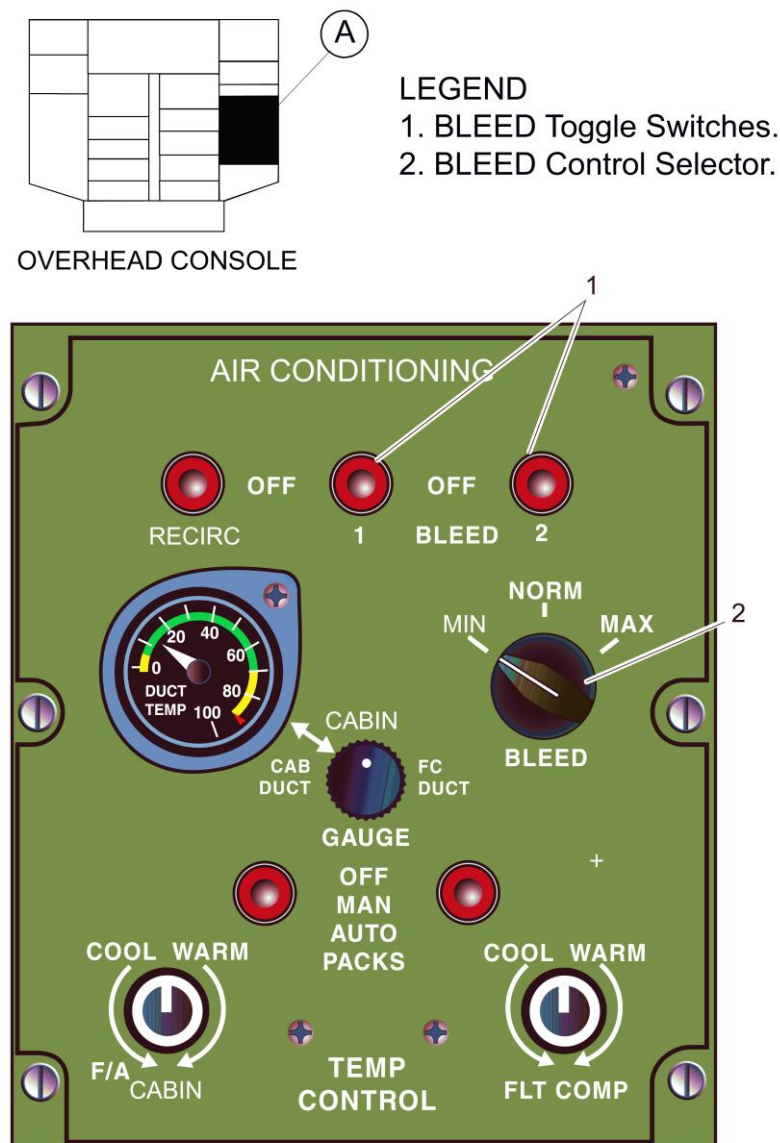


Figure 36-27 Bleed Air, Air Conditioning Control Panel

The Bleed Air System has the following caution lights:

- #1 BLEED HOT
- #2 BLEED HOT

[Refer Figure 36-28](#)

The #1 BLEED HOT caution light illuminates when:

- Left bleed overpressure switch detects a pressure of greater than 100 psig.
- Left bleed over-temperature switch detects a temperature of greater than 349°C (660°F).
- Left nacelle duct leak over-temperature switch detects a temperature of greater than 182°C (360°F).

The #2 BLEED HOT caution light illuminates on when:

- Right bleed overpressure switch detects a pressure of greater than 100 psig.
- Right bleed over-temperature switch detects a temperature of greater than 349°C (660°F).
- Right nacelle duct leak over-temperature switch detects a temperature of greater than 182°C (360°F).

Both #1 and #2 BLEED HOT caution lights illuminate when:

- Both aft duct leak switches detect a shroud temperature greater than 180°C.

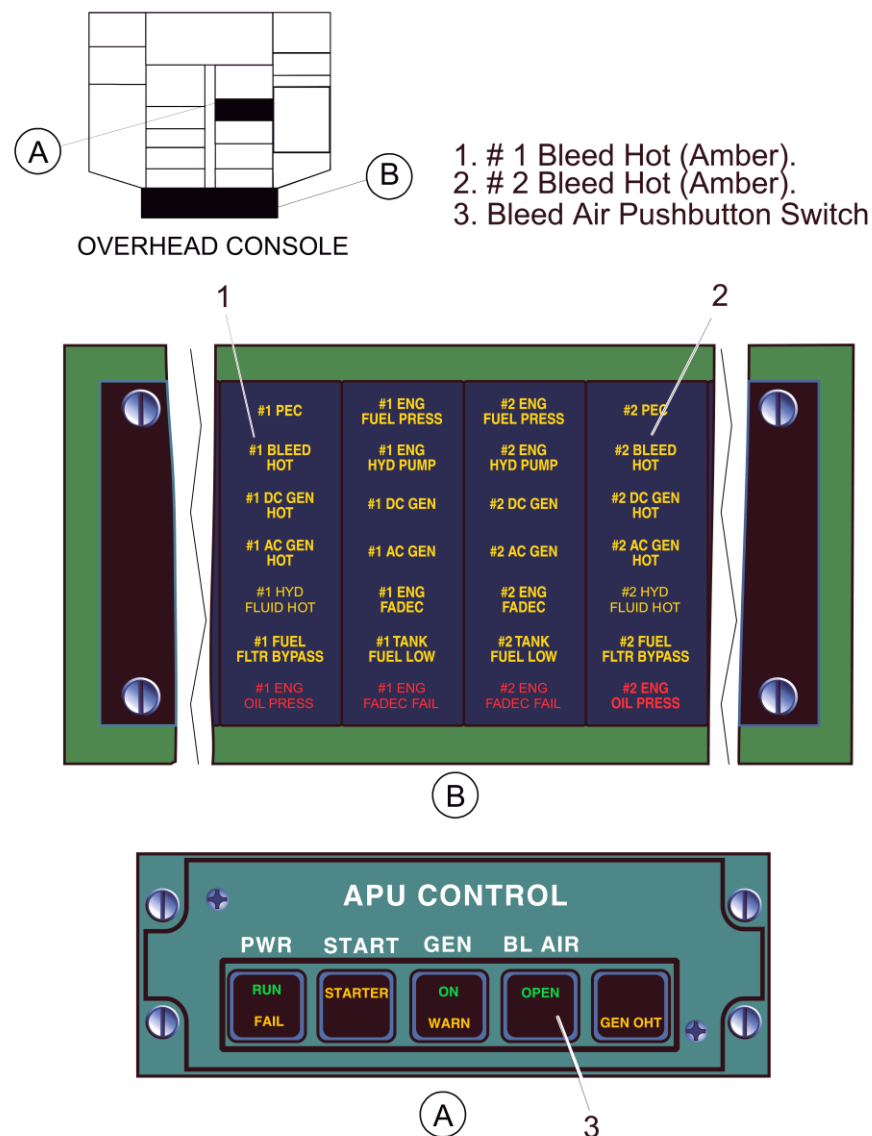


Figure 36-28 APU Control and Caution Panel

Engine Bleed Air

[Refer Figure 36-29\]](#)

- Selection of Bleed switch 1 provides a ground to close relay 2–K3 (Relay JB2). 28 VDC for the relay is supplied from the left and the right main DC Bus.
- A “Bleed Selected On” discrete signal from the ECU (Pin 36) is returned to the ECU (Pin 56) through relay contacts B1 and B2.
- The ECU outputs a signal to open the NSOV (Pins 53 and 54).
- Excitation voltage from the ECU (Pin 71) is sent to the rotary BLEED (MIN/NORM/MAX) switch and returned through pins 9, 36 or 57.
- This input along with differential pressure sensor input is used to adjust the current delivered to the NSOV for modulation.
- HPSOV torque motor control is through the normally closed contacts (A2 and B2) of relay 2-K2. Valve position feedback is provided through Pins 72 and 73.

Figure 36-29 Bleed ON & De-ice OFF Condition – Electrical Schematic

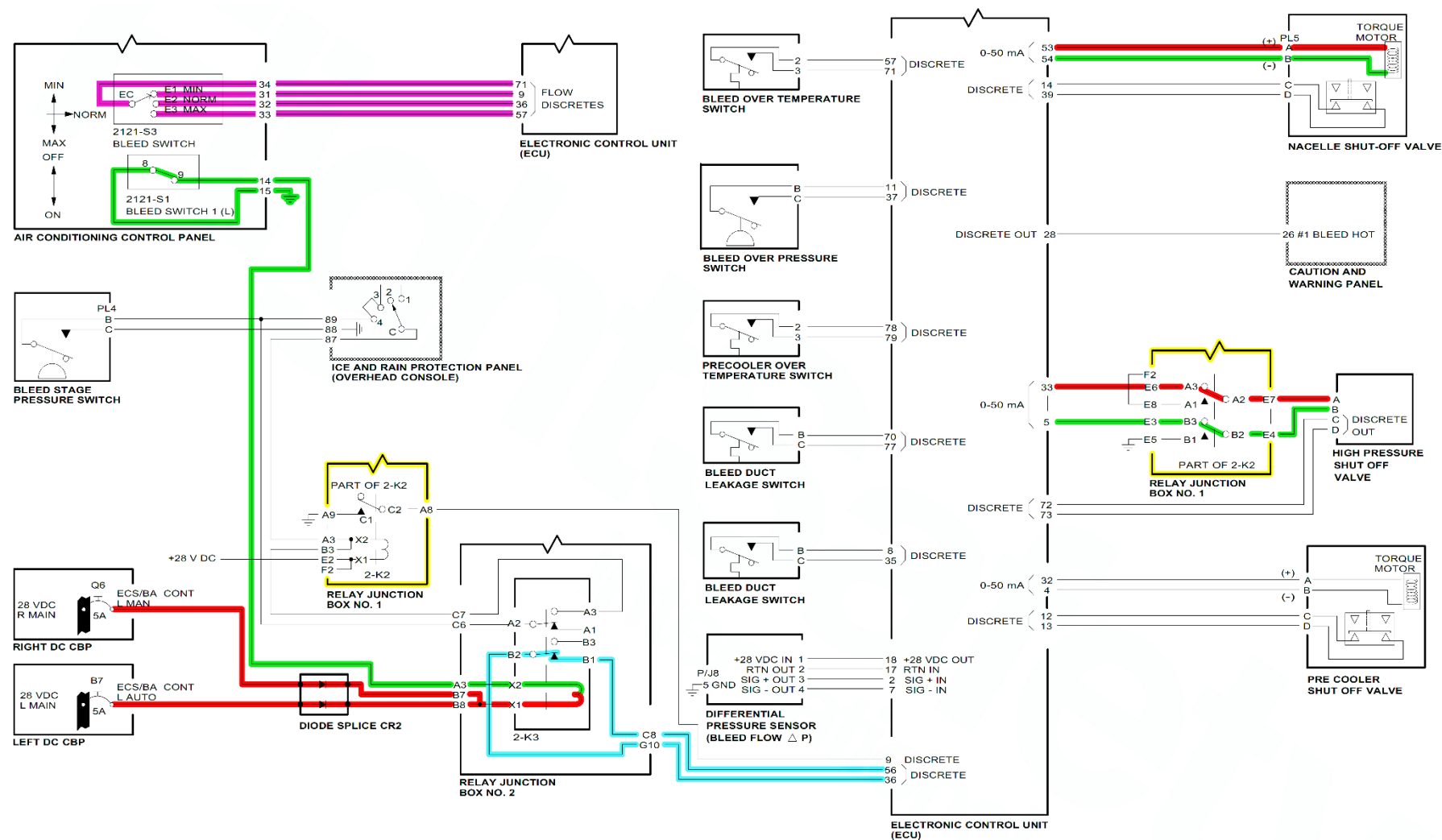


Figure 36-29 Bleed ON & De-ice OFF Condition – Electrical Schematic**BLEED OFF OR DE-ICE ON – HPSOV Control**

[Refer Figure 36-30](#)

- If the Bleed switch is selected off OR the airframe De-Ice is selected on, control of the HPSOV is via the Bleed Stage Pressure switch.
- Relay 2-K3 is energized and relay 2-K2 is relaxed. When the P3 Bleed stage pressure switch senses pressure less than 77 psi (531 kPa), it closes, supplying a ground to energize relay 2-K2.
- Energizing 2-K2 removes control of the HPSOV from the ECU and contacts A2 and B2 supply 28 VDC and a ground to open the HPSOV.

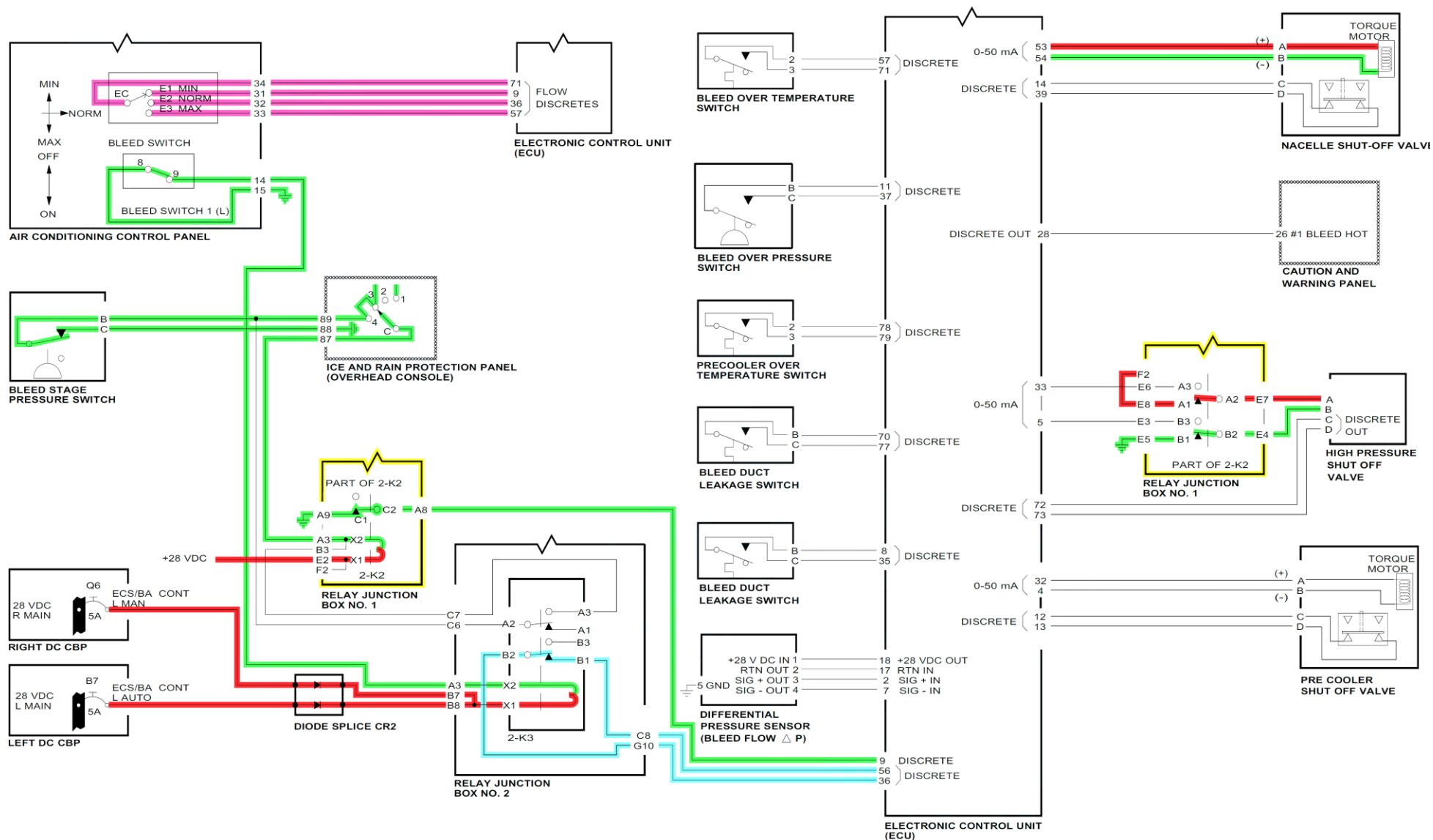


Figure 36-30 Bleed ON & De-ice ON Condition – Electrical Schematic

Control and Fault

[Refer Figure 36-31](#)

Control

When P3 air temperature is greater than 288°C (550°F), the precooler over-temperature switch closes and sends a discrete signal to the ECU (Pin 79). The ECU will command the P2.2 SOV open.

Fault

Bleed over-temperature switch senses temperature exceeding 349°C (660°F), a discrete signal is sent to the ECU Pin 71.

Bleed over-pressure switch senses a pressure in excess of 100 psig (689 kPa), a discrete signal is sent to the ECS ECU Pin 37.

Duct leak switch senses temperature in the shroud greater than 182°C (360°F), a discrete signal is sent to the ECS ECU Pin 77.

Upon receipt of any of the above discretes, the ECU will shut down the system by sending signals to close the HPSOV, NSOV and the P2.2 SOV. The ECU will output a signal to illuminate the associated BLEED HOT caution light and log a related fault code.

The system will not automatically reset. The bleed switches must be selected to OFF then ON to reset the system.

If either of the duct leak switches in the common shroud area (Aft Fuselage Area) senses a temperature exceeding 182°C (360°F), a discrete signal is sent to the ECSECU Pin 8 and to the APU relay panel Pin E-.

The ECU will send a signal to the APU FADEC for APU bleed valve control. The APU relay panel will interrupt the bleed request and the APU bleed valve will close.

The ECU will output a signal to illuminate both BLEED HOT caution lights. The system will not automatically reset. All bleed switches (Engine and APU) must be selected to OFF then ON to reset the system.

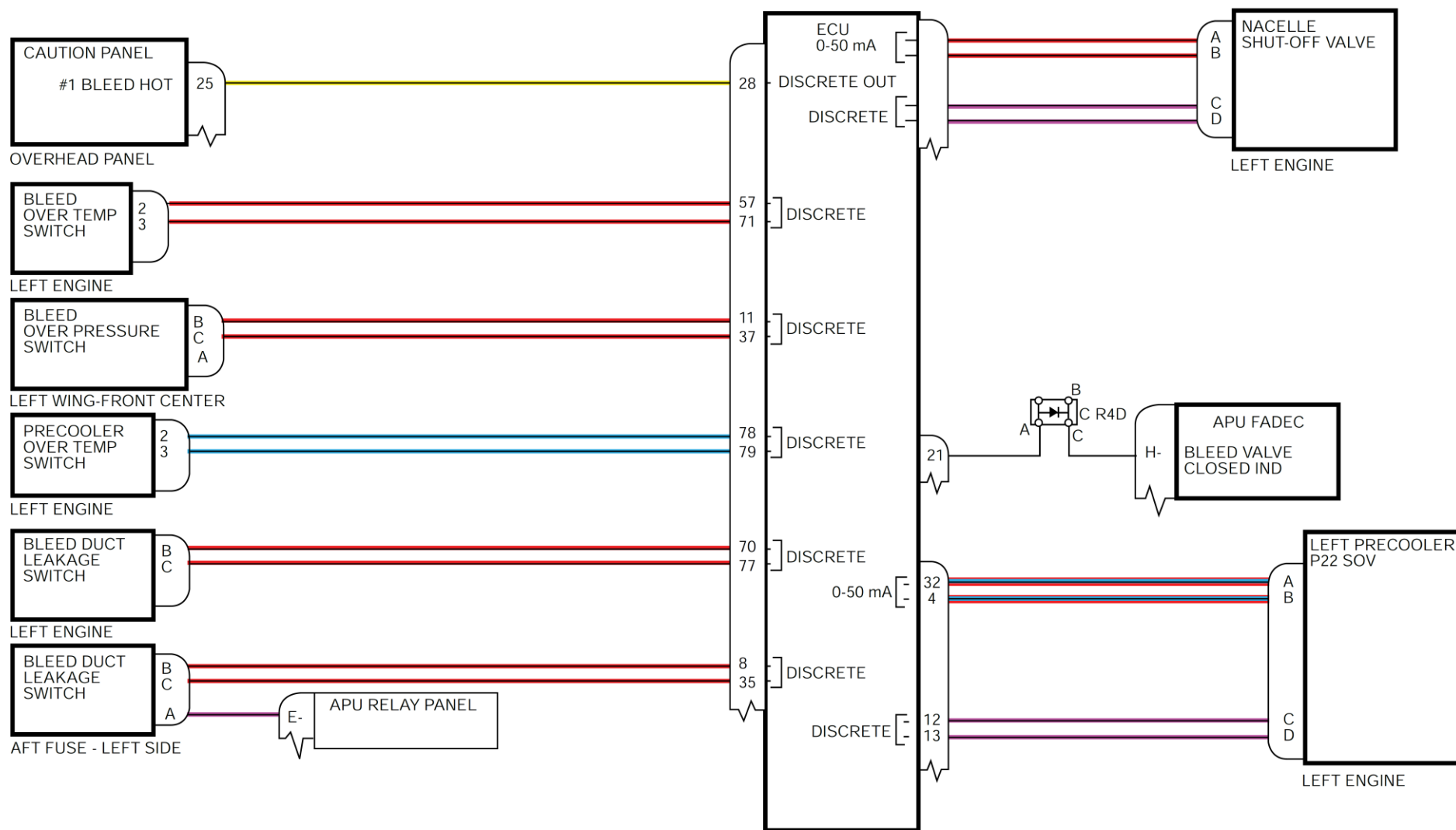


Figure 36-31 Bleed Air System – Control and fault

APU Bleed Air

Selection of the BL AIR switch on the APU CONTROL panel energizes the APU bleed air system. The APU will supply the ECS with bleed air only when the two engine BLEED 1 and 2 switches are selected off.

Deice and Oil Cooler Ejector Bleed Air

Bleed air from each engine is always available for the deice system and the oil cooler ejectors.

P2.7 bleed air flows through a smaller deice pre-cooler in the precooler reducing the deice bleed air temperature. This deice precooler uses a continuous flow of P2.2 bleed air to reduce the temperature of the P2.7 bleed air before it flows to the deice system.

The oil cooler ejector shutoff valve is opened by the FADEC when the conditions that follow occur:

- Aircraft WOW,
- PLA less 60 degrees
- Oil temperature is greater than 104°C